Math 230B Notes

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Chapter 1

Differentiation

1.1 The Derivative of a Function

Definition 1.1.1. (Differentiability and the Derivative)
Let $I \subseteq \mathbb{R}$ be an interval, $f : I \to \mathbb{R}$, and $c \in I$.

(*i*) We say f is differentiable at c if

$$\lim_{x \to c} \frac{f(x) - f(c)}{x - c}$$

exists (that is, it equals a real number). In this case, the quantity $\lim_{x \to c} \frac{f(x)-f(c)}{x-c}$ is called the derivative of f at c and is denoted by

$$f'(c), \quad \frac{df}{dx}(c), \quad \frac{df}{dx}|_{x=c}$$

(*ii*) If $f : I \to \mathbb{R}$ is differentiable at every point $c \in I$, we say f is differentiable (on I).

Remark 1.1.1. Note that

$$\begin{aligned}
f'(c) = L &\iff \lim_{x \to c} \frac{f(x) - f(c)}{x - c} = L \\
&\iff \forall \epsilon > 0 \ \exists \delta > 0 \text{ such that if } 0 < |x - c| < \delta, \text{ then } \left| \frac{f(x) - f(c)}{x - c} - L \right| < \epsilon \\
&\iff \forall \epsilon > 0 \ \exists \delta > 0 \text{ such that if } 0 < |h < \delta, \text{ then } \left| \frac{f(c + h) - f(c)}{h} - L \right| < \epsilon \quad \text{(Let } h = x - c\text{)} \\
&\iff \lim_{h \to 0} \frac{f(c + h) - f(c)}{h} = L
\end{aligned}$$

Remark 1.1.2. Let A denote the collection of all points at which $f : I \to \mathbb{R}$ is differentiable. If $A \neq \emptyset$, the function $f' : A \to \mathbb{R}$ defined by

$$f'(c) = \lim_{x \to c} \frac{f(x) - f(c)}{x - c} \quad \forall c \in A$$

is called the derivative of f.

Example 1.1.1. Let $I \subseteq \mathbb{R}$ be an interval and $f : I \to \mathbb{R}$ be given by $f(x) = x^2$. Prove that f is differentiable on I and find the derivative.

Proof. $\forall c \in I$,

$$\begin{aligned}
\lim_{x \to c} \frac{f(x) - f(c)}{x - c} &= \lim_{x \to c} \frac{x^2 - c^2}{x - c} \\
&= \lim_{x \to c} x + c \\
&= 2c \qquad \text{(is continuous)}
\end{aligned}$$

So, $\forall c \in I \quad f'(c) = 2c$. Hence,

$$f' : I \to \mathbb{R}, \quad f'(x) = 2x.$$

□

Example 1.1.2. Let $I \subseteq \mathbb{R}$ be an interval and $f : I \to \mathbb{R}$ be given by $f(x) = x^n$ where $n \in \mathbb{N}, \quad n \geq 3$. Prove that f is differentiable on I and find the derivative.

Proof.

$$\begin{aligned}
\lim_{x\to c} \frac{f(x) - f(c)}{x - c} &= \lim_{x\to c} \frac{x^n - c^n}{x - c} \\
&= \lim_{x\to c} \frac{(x - c)(x^{n-1} + cx^{n-2} + ... + c^{n-1})}{x - c} && \text{(Algebra)} \\
&= \lim_{x\to c} \left[x^{n-1} + cx^{n-1} + ... + c^{n-1}\right] \\
&= c^{n-1} + c \cdot c^{n-2} + ... + c^{n-1} && \text{(Continuity)} \\
&= n \cdot c^{n-1}
\end{aligned}$$

So, $\forall c \in I \quad f'(c) = n \cdot c^{n-1}$. Hence,

$$f' : I \to \mathbb{R}, \quad f'(x) = nx^{n-1}.$$

□

Example 1.1.3. Prove that $f : \mathbb{R} \to \mathbb{R}, \quad f(x) = |x|$ is not differentiable at $c = 0$.

Proof. We need to show that $\lim_{x\to c} \frac{f(x)-f(0)}{x-0}$ does not exist. Note that

$$\frac{f(x) - f(0)}{x - 0} = \frac{|x| - |0|}{x - 0} = \frac{|x|}{x}$$

Let $g(x) = \frac{|x|}{x}$. We want to show $\lim_{x\to 0} g(x)$ does not exist. By the sequential criterion for limits of functions, it is enough to find two sequences (a_n) and (b_n) in $\mathbb{R}\backslash\{0\}$ such that $a_n \to 0$ and $b_n \to 0$, but $\lim g(a_n) \neq \lim g(b_n)$. Let $a_n = -\frac{1}{n}$ and $b_n = \frac{1}{n}$. Clearly, $a_n \to 0$ and $b_n \to 0$. However,

$$\lim_{n\to\infty} g(a_n) = \lim_{n\to\infty} \frac{|a_n|}{a_n} = \lim_{n\to\infty} \frac{|-1/n|}{-1/n} = \lim_{n\to\infty} (-1) = -1$$

$$\lim_{n\to\infty} g(b_n) = \lim_{n\to\infty} \frac{|b_n|}{b_n} = \lim_{n\to\infty} \frac{|1/n|}{1/n} = \lim_{n\to\infty} (1) = 1$$

□

Theorem 1.1.1. (Differentiable $\implies$ Continuous)
Let $I \subseteq \mathbb{R}$ be an interval, $c \in I$, and $f : I \to \mathbb{R}$ be differentiable at c. Then f is continuous at c.

Proof. It is enough to show that $\lim_{x\to c} f(x) = f(c)$ (an interval doesn't have an isolated point). Note that

$$\begin{aligned}
\lim_{x\to c} (f(x) - f(c)) &= \lim_{x\to c} \left[\frac{f(x) - f(c)}{x - c}(x - c)\right] \\
&= \left[\lim_{x\to c} \frac{f(x) - f(c)}{x - c}\right] \left[\lim_{x\to c}(x - c)\right] && \text{(ALT for Functions)} \\
&= f'(c) \cdot 0 = 0.
\end{aligned}$$

So,

$$\begin{aligned}\lim_{x\to c} f(x) &= \lim_{x\to c}[f(x) - f(c) + f(c)] \\ &= \lim_{x\to c}[f(x) - f(c)] + \lim_{x\to c} f(c) \\ &= 0 + f(c) \\ &= f(c).\end{aligned}$$

□

Corollary 1.1.1. If $f : I \to \mathbb{R}$ is not continuous at $c \in I$, then f is not differentiable at c.

Example 1.1.4. Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = \begin{cases} x^2 & \text{if } x \in \mathbb{Q} \\ 0 & \text{if } x \notin \mathbb{Q} \end{cases}$.

(*i*) Prove f is continuous at 0.

(*ii*) Prove f is discontinuous at all $x \neq 0$.

(*iii*) Prove that f is nondifferentiable at all $x \neq 0$.

(*iv*) Prove that $f'(0) = 0$.

Proof. (*i*) We need to show that

$$\forall \epsilon > 0\ \exists \delta > 0 \text{ such that if } |x - 0| < \delta \text{ then } |f(x) - f(0)| < \epsilon$$

Let $\epsilon > 0$ be given. Our goal is to find $\delta > 0$ such that

$$\text{if } |x| < \delta \text{ then } |f(x)| < \epsilon \qquad (*)$$

Informal Discussion

Note that

Case 1: if $x \notin \mathbb{Q}$ then $|f(x)| = |0| < \epsilon$ ✓

Case 2: if $x \in \mathbb{Q}$ then $|f(x)| = |x^2| = |x|^2$

So, we want to find δ such that if $|x| < \delta$, then $|x|^2 < \epsilon$. Clearly, $\delta = \sqrt{\epsilon}$ works.

We claim that $(*)$ holds with $\delta = \sqrt{\epsilon}$. See the discussion.

(*ii*) Let $c \neq 0$. Our goal is to show f is discontinuous at c. By the sequential criterion for continuity, it is enough to find a sequence (a_n) such that $a_n \to c$ but $f(a_n) \not\to f(c)$. We proceed by two cases:

Case 1: $c \notin \mathbb{Q}$

$\mathbb{Q}$ is dense in $\mathbb{R}$, so there exists a sequence of rational numbers (r_n) such that $r_n \to c$. We have

$$\left.\begin{array}{l} f(r_n) = r_n^2\ \forall n \\ f(c) = 0 \end{array}\right\} \implies f(r_n) \not\to f(c)$$

$$\left.\begin{array}{l} r_n \to c \\ f(r_n) \not\to f(c) \end{array}\right\} \implies f \text{ is discontinuous at } c.$$

(*iii*) Let $c \neq 0$. By (*ii*), f is not continuous at c. Therefore, f is not differentiable at c.

(*iv*) We need to show $\lim_{x\to c} \frac{f(x)-f(0)}{x-0} = 0$. Note that

$$\frac{f(x) - f(0)}{x - 0} = \frac{f(x)}{x}.$$

Our goal is to show:

$$\forall \epsilon > 0\ \exists \delta > 0 \text{ such that if } 0 < |x - 0| < \delta \text{ then } \left|\frac{f(x)}{x} - 0\right| < \epsilon.$$

Let $\epsilon > 0$ be given. Our goal is to find $\delta > 0$ such that

$$\text{if } 0 < |x| < \delta, \text{ then } \left| \frac{f(x)}{x} - 0 \right| < \epsilon \qquad (*)$$

We claim that $(*)$ holds with $\delta = \epsilon$ (or any postive number less than ϵ). Indeed, if $x \in \mathbb{R}$ such that $0 < |x| < \delta = \epsilon$, then

Case 1: $x \notin \mathbb{Q}$

$$\left| \frac{f(x)}{x} \right| = \left| \frac{0}{x} \right| = 0 < \epsilon.$$

Case 2: $x \in \mathbb{Q}$

$$\left| \frac{f(x)}{x} \right| = \left| \frac{x^2}{x} \right| = |x| < \delta = \epsilon.$$

□

Theorem 1.1.2. (Algebraic Differentiability Theorem)
Assume $f : I \to \mathbb{R}$ and $g : I \to \mathbb{R}$ are differentiable at $c \in I$. Then

(*i*) $\forall k \in \mathbb{R}$, kf is differentiable at c and

$$(kf)'(c) = k \cdot f'(x)$$

(*ii*) $f + g$ is differentiable at c and

$$(f + g)'(c) = f'(c) + g'(c)$$

(*iii*) fg is differentiable at c and

$$(fg)'(c) = f'(c)g(c) + f(c)g'(c)$$

(*iv*) $\frac{f}{g}$ is differentiable at c (provided $g(c) \neq 0$) and

$$\left(\frac{f}{g}\right)'(c) = \frac{f'(c)g(c) - f(c)g'(c)}{[g(c)]^2}$$

Proof. Here, we will prove (*ii*) and (*iii*).

(*ii*)

$$\begin{aligned}
\lim_{x \to c} \frac{(f+g)(x) - (f+g)(c)}{x - c} &= \lim_{x \to c} \frac{f(x) + g(x) - f(c) - g(c)}{x - c} \\
&= \lim_{x \to c} \frac{f(x) - f(c)}{x - c} + \lim_{x \to c} \frac{g(x) - g(c)}{x - c} \\
&= f'(c) + g'(c).
\end{aligned}$$

So, $f + g$ is differentiable at c, and $(f + g)'(c) = f'(c) + g'(c)$.

(*iiii*)

$$\begin{aligned}
\lim_{x \to c} \frac{(fg)(x) - (fg)(c)}{x - c} &= \lim_{x \to c} \frac{f(x)g(x) - f(c)g(c)}{x - c} \\
&= \lim_{x \to c} \frac{f(x)g(x) - f(c)g(x) + f(c)g(x) - f(c)g(c)}{x - c} \\
&= \lim_{x \to c} \frac{(f(x) - f(c))\, g(x) + f(c)\, (g(x) - g(c))}{x - c} \\
&= \left[\lim_{x \to c} \frac{f(x) - f(c)}{x - c}\right] \left[\lim_{x \to c} g(x)\right] + \left[\lim_{x \to c} f(c)\right] \left[\lim_{x \to c} \frac{g(x) - g(c)}{x - c}\right] \\
&= f'(c) \cdot g(c) + f(c) \cdot g'(c)
\end{aligned}$$

Thus fg is differentiable at c, and $(fg)'(c) = f'(c)g(c) + f(c)g'(c)$. □

Theorem 1.1.3. (Chain Rule)
Let $I_1 \subseteq \mathbb{R}$ and $I_2 \subseteq \mathbb{R}$ be two intervals. Suppose $f : I_1 \to \mathbb{R}$ and $g : I_2 \to \mathbb{R}$ such that $f(I_1)$ is contained in I_2, f is differentiable at $c \in I_2$, and g is differentiable at $f(c) \in I_2$. Then the function $g \circ f : I_1 \to \mathbb{R}$ is differentiable at $c \in I_1$, and

$$(g \circ f)'(c) = g'(f(c)) \cdot f'(c).$$

Informal Discussion

The following is an incorrect proof of the theorem:

$$\begin{aligned}
\lim_{x\to c} \frac{(g\circ f)(x) - (g\circ f)(c)}{x-c} &= \lim_{x\to c} \frac{g(f(x)) - g(f(c))}{x-c} \\
&= \lim_{x\to c} \frac{g(f(x)) - g(f(c))}{f(x)-f(c)} \cdot \frac{f(x)-f(c)}{x-c} \\
&= \left[\lim_{x\to c} \frac{g(f(x)) - g(f(c))}{f(x)-f(c)}\right] \cdot \left[\lim_{x\to c}\lim_{x\to c} \frac{f(x)-f(c)}{x-c}\right] \\
&= g'(f(c)) \cdot f'(c)
\end{aligned}$$

This proof fails because even though $x \to c \implies x \neq c$, it's not necessarily the case that $f(x) \to f(c) \implies f(x) \neq f(c)$. I.e., the algebraic limit theorem for functions fails as $f(x) - f(c)$ might be zero. Dividing by $f(x) - f(c)$ is not legitimate. To see why this fails, consider the case when f is a constant function. We instead use the following idea: Replace $\frac{g(f(x))-g(f(c))}{f(x)-f(c)}$ with a new function $d(f(x))$ such that

(i) $d(f(x)) = \frac{g(f(x))-g(f(c))}{f(x)-f(c)}$ when $f(x) \neq f(c)$

(ii) $d(f(x))$ is defined even when $f(x) = f(c)$

(iii) $d(f(x)) \cdot \frac{f(x)-f(c)}{x-c} = \frac{g(f(x))-g(f(c))}{x-c}$ for all $x \in I_1, x \neq c$

Proof. Let $d : I_2 \to \mathbb{R}$ be defined by

$$d(y) = \begin{cases} \frac{g(y)-g(f(c))}{y-f(c)} & y \neq f(c) \\ \lim\limits_{y\to f(c)} \frac{g(y)-g(f(c))}{y-f(c)} = g'(f(c)) & y = f(c) \end{cases}$$

Clearly, d satisfies requirements (i) and (ii) from above.

Observation 1: d is continuous at $f(c)$. Indeed,

$$\lim_{y\to f(c)} d(y) = \lim_{y\to f(c)} \frac{g(y) - g(f(c))}{y - f(c)} = d(f(c))$$

Observation 2: For all $x \in I_1$ with $x \neq c$, we have

$$d(f(x)) \cdot \frac{f(x) - f(c)}{x - c} = \frac{g(f(x)) - g(f(c))}{x - c}$$

This is true because

Case 1: $f(x) \neq f(c)$

$$\begin{aligned}
d(f(x)) \cdot \frac{f(x) - f(c)}{x - c} &= \frac{g(f(x)) - g(f(c))}{f(x) - f(c)} \cdot \frac{f(x) - f(c)}{x - c} \\
&= \frac{g(f(x)) - g(f(c))}{x - c}
\end{aligned}$$

Case 2: $f(x) = f(c)$

$$\begin{aligned}
LHS = d(f(x)) \cdot \frac{f(x) - f(c)}{x - c} = d(f(c)) \cdot \frac{f(x) - f(c)}{x - c} = g'(f(c)) \cdot 0 = 0 \\
RHS = \frac{g(f(x)) - g(f(c))}{x - c} = \frac{g(f(c)) - g(f(c))}{x - c} = 0
\end{aligned}$$

So, $LHS = RHS = 0$.

We have,

$$\begin{aligned}\lim_{x\to c}\frac{(g\circ f)(x)-(g\circ f)(c)}{x-c} &= \lim_{x\to c}\frac{g(f(x))-g(f(c))}{x-c}\\ &= \lim_{x\to c}\left[d(f(x))\cdot\frac{f(x)-f(c)}{x-c}\right]\\ &= \left[\lim_{x\to c}(d\circ f)(x)\right]\cdot\left[\lim_{x\to c}\frac{f(x)-f(c)}{x-c}\right]\\ &\overset{(*)}{=} (d\circ f)(c)\cdot f'(c)\\ &= d(f(c))\cdot f'(c)\\ &= g'(f(c))\cdot f'(c)\end{aligned}$$

So, $g\circ f$ is differentiable at c and $(g\circ f)'(c) = g'(f(c))\cdot f'(c)$.

$(*)$ Note that f is continuous at c and d is continuous at $f(c)$, so by composition of continuos functions we conclude that $d\circ f$ is continuous at c and

$$\lim_{x\to c}(d\circ f)(c) = (d\circ f)(c).$$

□

Example 1.1.5. Let $f:\mathbb{R}\to\mathbb{R}$ be defined by $f(x) = \begin{cases} x^2\sin\frac{1}{x} & \text{if } x\neq 0\\ 0 & \text{if } x=0\end{cases}$

(*i*) Prove that f is differentiable at all $x\neq 0$.

(*ii*) Prove that $f'(0) = 0$

(*iii*) Prove that f' is not continuous at 0.

Proof. (*i*) We have

$$\left.\begin{array}{ll} h_1(x) = 1 & \text{is differentiable on } \mathbb{R}\\ h_2(x) = x & \text{is differentiable on } \mathbb{R}\end{array}\right\} \Longrightarrow \frac{h_1(x)}{h_2(x)} = \frac{1}{x} \text{ is differentiable at all } x\neq 0$$

$$\left.\begin{array}{ll} h_3(x) = \sin x & \text{is differentiable on } \mathbb{R}\\ h_4(x) = \frac{1}{x} & \text{is differentiable at all } x\neq 0\end{array}\right\} \Longrightarrow (h_3\circ h_4)(x) = \sin\frac{1}{x} \text{ is differentiable at all } x\neq 0$$

$$\left.\begin{array}{ll} h_5(x) = x^2 & \text{is differentiable on } \mathbb{R}\\ h_4(x) = \sin\frac{1}{x} & \text{is differentiable at all } x\neq 0\end{array}\right\} \Longrightarrow h_5(x)\cdot h_6(x) = x^2\sin\frac{1}{x} \text{ is differentiable at all } x\neq 0$$

Indeed, it follows from the algebraic differentiation theorem and the chain rule that

$$\begin{aligned}(x^2\sin\frac{1}{x})' &= (x^2)'\cdot\sin\frac{1}{x} + x^2\cdot(\sin\frac{1}{x})'\\ &= 2x\cdot\sin\frac{1}{x} + x^2\left[(\cos\frac{1}{x})(-\frac{1}{x^2})\right]\\ &= 2x\sin\frac{1}{x} - \cos\frac{1}{x}\end{aligned}$$

(*ii*) Note that $f(0) = 0$ does not imply $f'(0) = 0$. When we want to compute f' at any point, in particular at 0, we need to pay attention to the behavior of f in a neighborhood of the point and not just the value of the function at the point. The reason is that $f'(c)$ is defined by taking $\lim_{x\to c}$.

Our goal is to show

$$\lim_{x\to c}\frac{f(x)-f(0)}{x-0} = 0$$

Note that

$$\frac{f(x)-f(0)}{x-0} = \frac{x^2\sin\frac{1}{x}-0}{x-0} = x\sin\frac{1}{x}$$

We want to show

$$\lim_{x\to 0}\left(x\sin\frac{1}{x}\right)=0$$

We have,

$$\left.\begin{array}{l}0\le\left|x\sin\frac{1}{x}\right|\le|x|\\ \lim\limits_{x\to 0}0=0\\ \lim\limits_{x\to 0}|x|=|0|=0\end{array}\right\}\overset{\text{SQZ Thm}}{\Longrightarrow}\lim_{x\to 0}\left|x\sin\frac{1}{x}\right|=0$$

Thus $\lim\limits_{x\to 0}x\sin\frac{1}{x}=0$.

(*iii*) According to parts (*i*) and (*ii*):

$$f':\mathbb{R}\to\mathbb{R},\quad f(x)=\begin{cases}2x\sin\frac{1}{x}-\cos\frac{1}{x} & \text{if } x\neq 0\\ 0 & \text{if } x=0\end{cases}$$

By the sequential criterion for continuity, it is enough to find a sequence (a_n) such that

$$a_n\to 0 \text{ but } f'(a_n)\not\to f'(0)$$

Let $a_n=\frac{1}{2n\pi}$. Clearly, $a_n\to 0$. However,

$$\begin{aligned}\lim_{n\to\infty}f'(a_n)&=\lim_{n\to\infty}\left[\frac{2}{2n\pi}\sin(2n\pi)-\cos(2n\pi)\right]\\ &=0-1\\ &\neq 0.\end{aligned}$$

□

1.2 Local Extrema

Definition 1.2.1. (Local Maximum, Local Minimum)
Let $\emptyset \neq A \subseteq (X, d)$, and let $f : A \to \mathbb{R}$.

(*i*) We say that f has a local maximum at $c \in A$ if

$$\exists \delta > 0 \text{ such that } \forall x \in N_\delta(c) \cap A \quad f(x) \leq f(c)$$

(*ii*) We say that f has a local minimum at $c \in A$ if

$$\exists \delta > 0 \text{ such that } \forall x \in N_\delta(c) \cap A \quad f(x) \geq f(c)$$

Lemma 1.2.1. (Order Limit Theorem for Functions)
Suppose $\lim_{x\to c} g(x)$ and $\lim_{x\to c} h(x)$ both exist.

(*i*) If $\exists \delta > 0$ such that $\forall x \in (c-\delta, c) \quad h(x) \leq g(x)$, then $\lim_{x\to c} h(x) \leq \lim_{x\to c} g(x)$

(*ii*) If $\exists \delta > 0$ such that $\forall x \in (c, c+\delta) \quad h(x) \leq g(x)$, then $\lim_{x\to c} h(x) \leq \lim_{x\to c} g(x)$

Proof. Here we will prove (*i*). The proof of (*ii*) is analogous. Let (a_n) be a sequence in $(c-\delta, c)$ such that $a_n \to c$. By the sequential criterion for limits of functions we have

$$a_n \to c \implies \begin{cases} \lim_{n\to\infty} g(a_n) = \lim_{x\to c} g(x) \\ \lim_{n\to\infty} h(a_n) = \lim_{x\to c} h(x) \end{cases} \tag{I}$$

Also note that

$$\begin{aligned} \forall n \quad a_n \in (c-\delta, c) &\implies \forall n \quad h(a_n) \leq g(a_n) \\ &\overset{\text{OLTS}}{\implies} \lim_{n\to\infty} h(a_n) \leq \lim_{n\to\infty} g(a_n) \end{aligned} \tag{II}$$

It follows from $(I), (II)$ that $\lim_{x\to c} h(x) \leq \lim_{x\to c} g(x)$. □

Theorem 1.2.1. (Interior Extremum Theorem)
Let $I \subseteq \mathbb{R}$ be an interval, $f : I \to \mathbb{R}$ be a function and $c \in \overline{I}$. Suppose f is differentiable at c. Then

(*i*) If f has a local maximum at c, then $f'(c) = 0$

(*ii*) If f has a local minimum at c, then $f'(c) = 0$

Proof. Here, we will prove (*i*). The proof for (*ii*) is analogous. Suppose f has a local maximum at c.

(i) f has a local maximum at $c \implies \exists \delta_1$ such that $\forall x \in (c-\delta_1, c+\delta_1) \cap I \quad f(x) \leq f(c)$

(ii) c is an interior point of $I \implies \exists \delta_2$ such that $(c-\delta_2, c+\delta_2) \subseteq I$

So, if we let $\delta = \min\{\delta_1, \delta_2\}$, then

$$\forall x \in (c-\delta, c+\delta) \quad f(x) \leq f(c)$$

We have

(*I*) For all $x \in (c-\delta, c)$

$$\left.\begin{matrix} x - c < 0 \\ f(x) \leq f(c) \end{matrix}\right\} \implies \frac{f(x)-f(c)}{x-c} \geq 0$$

$$\overset{OLTF}{\implies} \lim_{x\to c} \frac{f(x)-f(c)}{x-c} \geq \lim_{x\to c} 0$$

$$\implies f'(c) \geq 0.$$

(II) For all $x \in (c, c+\delta)$

$$\left.\begin{array}{l} x-c>0 \\ f(x) \le f(c) \end{array}\right\} \implies \frac{f(x)-f(c)}{x-c} \le 0$$
$$\overset{OLTF}{\implies} \lim_{x\to c} \frac{f(x)-f(c)}{x-c} \le \lim_{x\to c} 0$$
$$\implies f'(c) \le 0.$$

It follows from $(I), (II)$ that $f'(c) = 0$. □

Remark 1.2.1. The following are three techniques that can be used in proving the existence of a solution:

(i) Suppose $h : [a, b] \to \mathbb{R}$ is continuous. Let α be a given real number. One way to show there exists a number c such that $h(c) = \alpha$ is as follows:

$$\text{Prove that } m \le \alpha \le M \text{ where } \begin{cases} m = \min\{h(x) : x \in [a,b]\} \\ M = \max\{h(x) : x \in [a,b]\} \end{cases}$$

(ii) Suppose $g : [a, b] \to \mathbb{R}$ is differentiable. One way to prove that there exists a number c such that $g'(c) = 0$ is as follows:

$$\text{Prove there is a point in } (a,b) \text{ at which } g \text{ has a local maximum or a local minimum}$$

(iii) Suppose $h : [a, b] \to \mathbb{R}$ is differentiable. Let α be a given real number. One way to prove that there exists a number c such that $h'(c) = \alpha$ is as follows:

$$\text{Define } g(x) = h(x) - \alpha x \text{ and prove that there is a point } c \text{ at which } g'(c) = 0$$

Theorem 1.2.2. (Darboux's Theorem)
Suppose $f : [a, b] \to \mathbb{R}$ is differentiable such that $f'(a) < f'(b)$ (or $f'(b) < f'(a)$), and let $\alpha \in \mathbb{R}$ be such that $f'(a) < \alpha < f'(b)$ (or $f'(b) < \alpha < f'(a)$). Then

$$\exists c \in (a,b) \text{ such that } f'(c) = \alpha$$

Proof. Let $g : [a, b] \to \mathbb{R}$ be defined by $g(x) = f(x) - \alpha x$. It follows from the algebraic differentiability theorem that g is differentiable on $[a, b]$, and so it is continuous on $[a, b]$. It is enough to show that

$$\exists c \in (a,b) \text{ such that } g'(c) = 0$$

To this end, it is enough to show that $\exists c \in (a, b)$ at which g has a local minimum. We have

$$\left.\begin{array}{l} g \text{ is continuous on [a,b]} \\ [a,b] \text{ is compact} \end{array}\right\} \implies g \text{ attains its minimum on } [a,b]$$

Let $\hat{c}$ be a point at which g attains a minimum. In what follows we will show that $\hat{c} \in (a, b)$ and so it can be used as the c that we were looking for. Note that (since $g'(x) = f'(x) - \alpha$)

$$g'(a) = f'(a) - \alpha < 0$$
$$g'(b) = f'(b) - \alpha > 0$$

Claim 1: $\hat{c} \neq a$

Assume for contradiction that $\hat{c} = a$. Then

$$\forall x \in [a,b] \quad g(x) \ge g(a)$$

so,

$$\forall x \in [a,b] \quad \begin{cases} g(x) - g(a) \ge 0 \\ x - a > 0 \end{cases}$$

Thus

$$\forall x \in (a,b) \quad \frac{g(x)-g(a)}{x-a} \ge 0$$

Thus

$$\lim_{x\to c}\frac{g(x)-g(a)}{x-a}\geq\lim_{x\to a}0$$

That is, $g'(a)\geq 0$. This contradicts the fact that $g'(a)<0$.

Claim 2: $\hat{c}\neq b$

Assume for contradiction that $\hat{c}=b$. In a similar manner to claim 1:

$$\forall x\in[a,b]\quad g(x)\geq g(b)\implies\forall x\in[a,b]\quad\begin{cases}g(x)-g(b)\geq 0\\ x-b<0\end{cases}$$
$$\implies\forall x\in[a,b]\quad\frac{g(x)-g(b)}{x-b}\leq 0$$

Thus,

$$\lim_{x\to c}\frac{g(x)-g(b)}{x-b}\leq\lim_{x\to b}0$$

That is,

$$g'(b)\leq 0.$$

This contradicts the fact that $g'(b)>0$.

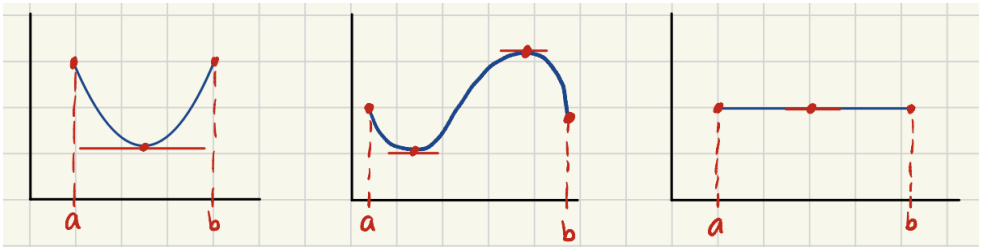
Example 1.2.1. Does there exist a differentiable function $f:[-1,1]\to\mathbb{R}$ whose derivative is $H:[-1,1]\to\mathbb{R}$ defined by

$$H(x)=\begin{cases}1 & 0\leq x\leq 1\\ 0 & -1\leq x\leq 0\end{cases}\ ?$$

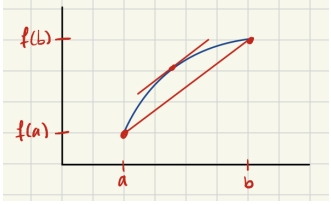
No! H does not have the intermediate value property. So, it cannot be the derivative of any differentiable function.

The following are some geometric conjectures involving the derivative of a function.

Conjecture 1.2.1.
Suppose $f:[a,b]\to\mathbb{R}$ is differentiable. Suppose $f(a)=f(b)$. Then there exists a point $c\in(a,b)$ at which the tangent line is horizontal. I.e., there exists $c\in(a,b)$ such that $f'(c)=0$.



Conjecture 1.2.2.
Suppose $f:[a,b]\to\mathbb{R}$ is differentiable. Then there exists a point $c\in(a,b)$ at which the tangent line is parallel to the line through the endpoints $(a,f(a))$ and $(b,f(b))$. I.e., there exists $c\in(a,b)$ such that $f'(c)=\frac{f(b)-f(a)}{b-a}$.



Conjecture 1.2.3.
Suppose $\vec{r}:[a,b]\to\mathbb{R}^2$, $\vec{r}(t)=(f(t),g(t))$ is a differentiable path in $\mathbb{R}^2$. Then there exists a point $\vec{r}(c)$ on the curve at which the tangent line is parallel to the line through the endpoints $\vec{r}(a)$ and $\vec{r}(b)$. Let's

try to find a mathematical formula for this statement:

$*$) The direction vector for the tangent line at the point $\vec{r}(c)$: $\vec{r}'(c) = (f'(c), g'(c))$

$*$) The direction vector for the line through the endpoints: $(f(b) - f(a), g(b) - g(a))$

So, assuming these vectors are nonzero, the claim of the conjecture can be described mathematically as

$$\exists c \exists \lambda \in \mathbb{R}\backslash\{0\} \text{ such that } (f'(c), g'(c)) = \lambda\,(f(b) - f(a), g(b) - g(a))$$

Note that

$$\begin{aligned}
(f'(c), g'(c)) = \lambda\,(f(b) - f(a), g(b) - g(a)) & \\
\implies & \begin{cases} f'(c) = \lambda\,(f(b) - f(a)) \\ g'(c) = \lambda\,(g(b) - g(a)) \end{cases} \\
\implies & \lambda f'(c)\,[g(b) - g(a)] = \lambda g'(c)\,[f(b) - f(a)] \\
\implies & f'(c)\,[f(b) - f(a)] = g'(c)\,[g(b) - g(a)]
\end{aligned}$$

1.3 Mean Value Theorems

We now study three theorems that make the previous geometric observations precise.

Theorem 1.3.1. (Rolle's Theorem)
Let $f : [a, b] \to \mathbb{R}$ be continuous. Let f be differentiable on (a, b). Suppose $f(a) = f(b)$. Then there exists a point $c \in (a, b)$ such that $f'(c) = 0$.

Proof. It is enough to show that there exists a point $c \in (a, c)$ at which f has a local maximum or a local minimum. We have

$$\left.\begin{array}{l} f \text{ is continuous} \\ [a,b] \text{ is compact} \end{array}\right\} \overset{EVT}{\Longrightarrow} f \text{ attains its maximum and minimum on } [a,b]$$

We consider two cases:

Case 1: Both $\max\limits_{a\le x\le b} f(x)$ and $\min\limits_{a\le x\le b} f(x)$ occur at the endpoints.
In this case, it follows from the assumption $f(a) = f(b)$ that $\max\limits_{a\le x\le b} f(x) = \min\limits_{a\le x\le b} f(x)$. So, f is a constant function on $[a, b]$. Hence

$$\forall x \in [a,b] \quad f'(x) = 0$$

So, we may choose c to be any point we like in (a, b).

Case 2: Either $\max\limits_{a\le x\le b} f(x)$ or $\min\limits_{a\le x\le b} f(x)$ occurs at a point $c \in (a, b)$.
It follows from the interior extreme value theorem that $f'(c) = 0$.

□

Theorem 1.3.2. (Mean Value Theorem)
Let $f : [a, b] \to \mathbb{R}$ be continuous and let f be differentiable on (a, b). Then there exists $c \in (a, b)$ such that $f'(c) = \frac{f(b)-f(a)}{b-a}$.

Proof. Let $g : [a, b] \to \mathbb{R}$ be defined by

$$g(x) = f(x) - \frac{f(b) - f(a)}{b - a}x$$

Note that

$*$) Algebraic continuity theorem $\implies$ g is continuous on $[a, b]$

$*$) Algebraic differentiability theorem $\implies$ g is differentiable on (a, b)

$*$) $g(a) = f(a) - \frac{f(b)-f(a)}{b-a}a = \frac{bf(a)-af(a)-af(b)+af(a)}{b-a} = \frac{bf(a)-af(a)}{b-a}$

$*$) $g(b) = f(b) - \frac{f(b)-f(a)}{b-a}b = \frac{bf(a)-af(b)}{b-a}$

g is continuous on $[a, b]$, differentiable on (a, b), and $g(a) = g(b)$. By Rolle's theorem,

$$\exists c \in (a,b) \text{ such that } g'(c) = 0$$

Note that $g'(x) = f'(x) - \frac{f(b)-f(a)}{b-a}$, so

$$\begin{aligned} g'(c) = 0 &\iff f'(c) - \frac{f(b) - f(a)}{b - a} = 0 \\ &\iff f'(c) = \frac{f(b) - f(a)}{b - a} \end{aligned}$$

□

Theorem 1.3.3. (Generalized Mean Value Theorem)
Let $f : [a, b] \to \mathbb{R}$ and $g : [a, b] \to \mathbb{R}$ be continuous functions that are differentiable on (a, b). Then there

exists a point $c \in (a,b)$ such that

$$f'(c)\,[g(b)-g(a)] = g'(c)\,[f(b)-f(a)]$$

Proof. Let $h(x) = [f(b)-f(a)]\,g(x) - [g(b)-g(a)]\,f(x)$. It follows from the assumptions, the algebraic continuity theorem, and the algebraic differentiability theorem that h is continuous on $[a,b]$ and differentiable on (a,b). Therefore, by the mean value theorem,

$$\exists c \in (a,b) \text{ such that } h'(c) = \frac{h(b)-h(a)}{b-a} \qquad (*)$$

Note that

$$\begin{aligned} h(b) &= [f(b)-f(a)]\,g(b) - [g(b)-g(a)]\,f(b) \\ &= f(b)g(b) - f(a)g(b) - g(b)f(b) + g(a)f(b) \\ &= g(a)f(b) - f(a)g(b) \\ h(a) &= [f(b)-f(a)]\,g(a) - [g(b)-g(a)]\,f(a) \\ &= f(b)g(a) - f(a)g(a) - g(b)f(a) + g(a)f(a) \\ &= f(b)g(a) - g(b)f(a) \end{aligned}$$

So $h(a) = h(b)$. Hence it follows from $(*)$ that $\exists c \in (a,b)$ such that $h'(c) = 0$ Now note that

$$\begin{aligned} h'(x) &= [f(b)-f(a)]\,g'(x) - [g(b)-g(a)]\,f'(x) \\ \implies h'(c) &= [f(b)-f(a)]\,g'(c) - [g(b)-g(a)]\,f'(c) \end{aligned}$$

Therefore,

$$\exists c \in (a,b) \text{ such that } [f(b)-f(a)]\,g'(c) - [g(b)-g(a)]\,f'(c) = 0$$

That is,

$$\exists c \in (a,b) \text{ such that } [f(b)-f(a)]\,g'(c) = [g(b)-g(a)]\,f'(c)$$

□

Remark 1.3.1. If g' is neer zero in (a,b), then we may rewrite the claim of general mean value theorem as follows:

$$\exists c \in (a,b) \text{ such that } \frac{f'(c)}{g'(c)} = \frac{f(b)-f(a)}{g(b)-g(a)}$$

Theorem 1.3.4.
Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$ be differentiable such that $f'(x) = 0\ \forall x \in I$. Then f is a constant function on I, that is, there exists $k \in \mathbb{R}$ such that $\forall x \in I, f(x) = k$.

Proof. Let $x, y \in I$ with $x < y$. It is enough to show that $f(x) = f(y)$. To this end, we wil apply the mean value theorem to f on the interval $[x,y]$:

$$\begin{aligned} \exists c \in (x,y) \text{ such that } f'(c) = \frac{f(y)-f(x)}{y-x} & \\ \implies 0 = \frac{f(y)-f(x)}{y-x} & \\ \implies 0 = f(y) - f(x) & \\ \implies f(x) = f(y) & \end{aligned}$$

□

Remark 1.3.2. Consider $f : A \to \mathbb{R}$ where $A = (-1,0) \cup (2,3)$ and $f(x) = \begin{cases} 1 & x \in (-1,0) \\ -1 & x \in (2,3) \end{cases}$ Then $\forall x \in A\ \ f'(x) = 0$, but f is not a constant function on A. The theorem above doesn't apply since A is not an interval.

Theorem 1.3.5.
Let $I \subseteq \mathbb{R}$ be an interval, and let $f : I \to \mathbb{R}$ and $g : I \to \mathbb{R}$ be differentiable such that $f'(x) = g'(x)\ \forall x \in I$. Then there exists $k \in \mathbb{R}$ such that $\forall x \in I,\ f(x) = g(x) + k$.

Proof. Let $h = f - g$. We have

$$\begin{aligned} &\forall x \in I \quad h'(x) = (f-g)'(x) = f'(x) - g'(x) = 0 \\ &\overset{1.3.4}{\implies} \exists k \in \mathbb{R} \text{ such that } \forall x \in I \quad h(x) = k \\ &\implies \exists k \in \mathbb{R} \text{ such that } \forall x \in I \quad f(x) - g(x) = k \\ &\implies \exists k \in \mathbb{R} \text{ such that } \forall x \in I \quad f(x) = g(x) + k \end{aligned}$$

□

Theorem 1.3.6.
Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$ be differentiable. Then

(i) f is increasing $\iff \forall c \in I \quad f'(c) \geq 0$

(ii) f is decreasing $\iff \forall c \in I \quad f'(c) \leq 0$

Proof.
Here, we will prove (i). The proof of (ii) is analogous.
$(\implies)$ Suppose f is increasing on I. Let $c \in I$. Note that for all $x \in I, x \neq c$ we have $\frac{f(x)-f(c)}{x-c} \geq 0$. Indeed,

$$\text{if } x > c \text{ then } \begin{cases} x - c > 0 \\ f(x) \geq f(c) \end{cases} \implies \frac{f(x)-f(c)}{x-c} \geq 0$$

$$\text{if } x < c \text{ then } \begin{cases} x - c < 0 \\ f(x) \leq f(c) \end{cases} \implies \frac{f(x)-f(c)}{x-c} \geq 0$$

It follows from the order limit theorem for functions that

$$\lim_{x \to c} \frac{f(x)-f(c)}{x-c} \geq \lim_{x \to c} 0$$

Hence, $f'(c) \geq 0$ as desired.

$(\impliedby)$ Suppose $\forall c \in I \quad f'(c) \geq 0$. Let $x_1, x_2 \in I$ with $x_1 < x_2$. It is enough to show that $f(x_1) \leq f(x_2)$. To this end, we apply the mean value theorem to the function f on $[x_1, x_2]$:

$$\exists c \in (x_1, x_2) \text{ such that } f'(c) = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$

So, $f(x_2) - f(x_1) = f'(c)(x_2 - x_1)$. Thus $f(x_2) - f(x_1) \geq 0$, that is, $f(x_1) \leq f(x_2)$ as desired. □

Theorem 1.3.7. (L'Hĉpital's Rule)
Let $I \subseteq \mathbb{R}$ be an interval, and $a \in I$. Let $f : [a, b] \to \mathbb{R}$ and $g : [a, b] \to \mathbb{R}$ be continuous. Suppose f and g are differentiable at all points in $I \backslash \{a\}$ and $f(a) = g(a) = 0$, $g'(x) \neq 0\ \forall x \in I \backslash \{a\}$ and $\lim_{x \to a} \frac{f'(x)}{g'(x)} = L \in \mathbb{R}$. Then $\lim_{x \to a} \frac{f(x)}{g(x)} = L$.

Proof. Our goal is to show that

$$\forall \epsilon > 0\ \exists \delta > 0 \text{ such that if } 0 < |x - a| < \delta \text{ (with } x \in I) \text{ then } \left| \frac{f(x)}{g(x)} - L \right| < \epsilon$$

Let $\epsilon > 0$. Our goal is to find $\delta > 0$ such that

$$\text{if } 0 < |x - a| < \delta \text{ (with } x \in I) \text{ then } \left| \frac{f(x)}{g(x)} - L \right| < \epsilon \qquad (*)$$

Since by assumption $\lim_{x\to a}\frac{f'(x)}{g'(x)} = L$, for the gien $\epsilon > 0$, there exists $\hat{\delta} > 0$ such that

$$\text{if } 0 < |x-a| < \hat{\delta} \text{ (with } x \in I) \text{ then } \left|\frac{f'(x)}{g'(x)} - L\right| < \epsilon$$

We claim that this $\hat{\delta}$ satisfies $(*)$. The reason is as follows:
Suppose $x \in I$ such that $0 < |x-a| < \hat{\delta}$. In what follows we will show that $\left|\frac{f(x)}{g(x)} - L\right| < \epsilon$. We consider two cases:

Case 1: $x > a \quad \left(x \in (a, a+\hat{\delta})\right)$

We apply the general mean value theorem to f and g on the interval $[a, x]$:

$$\exists c \in (a,x) \text{ such that } \frac{f'(x)}{g'(x)} = \frac{f(x)-f(a)}{g(x)-g(a)}$$

Since $f(a) = g(a) = 0$, we conclude that

$$\frac{f'(c)}{g'(c)} = \frac{f(x)}{g(x)}$$

It follows that

$$\left|\frac{f(x)}{g(x)} - L\right| = \left|\frac{f'(x)}{g'(x)} - L\right| < \epsilon$$

(the latter inequality is true because $0 < |c-a| \leq |x-a| \leq \hat{\delta}$)

Case 2: $x < a \quad \left(x \in (a-\hat{\delta}, a)\right)$

We apply the general mean value theorem to f and g on $[x, a]$:

$$\exists c \in (x,a) \text{ such that } \frac{f'(c)}{g'(c)} = \frac{f(a)-f(x)}{g(a)-g(x)}$$

Since $f(a) = g(a) = 0$, we conclude that

$$\frac{f'(c)}{g'(c)} = \frac{f(x)}{g(x)}$$

It follows that

$$\left|\frac{f(x)}{g(x)} - L\right| = \left|\frac{f'(x)}{g'(x)} - L\right| < \epsilon$$

(the latter inequality is true because $0 < |c-a| \leq |x-a| \leq \hat{\delta}$)

□

1.4 Taylor Polynomials

Consider $f : I \to \mathbb{R}$ given by $f(x) = (x - x_0)^k$. What do the n derivatives look like? If $n = 0$, then by the mean value theorem we have

$$\left.\begin{array}{l} f : I \to \mathbb{R} \text{ is differentiable} \\ f(x_0) = 0 \end{array}\right\} \implies f(x) = f'(c)(x - x_0)$$

Observation 1. Let k be a natural number. Let x_0 be a fixed number.

$*)$ $\frac{d}{dx}\left[(x-x_0)^k\right] = k(x-x_0)^{k-1}$

$*)$ $\frac{d^2}{dx^2}\left[(x-x_0)^k\right] = \frac{d}{dx}\left[k(x-x_0)^{k-1}\right] = k(k-1)(x-x_0)^{k-2}$

$*)$ $\frac{d^3}{dx^3}\left[(x-x_0)^k\right] = \frac{d}{dx}\left[k(k-1)(x-x_0)^{k-2}\right] = k(k-1)(k-2)(x-x_0)^{k-3}$

$\vdots$

$*)$ $\frac{d^k}{dx^k}\left[(x-x_0)^k\right] = k(k-1)\dots(2)(1)(x-x_0)^{k-k} = k!$

If $j < k$

$*)$ $\frac{d^j}{dx^j}\left[(x-x_0)^k\right] = k(k-1)\dots(k-(j-1))\,(x-x_0)^{k-j}$

Thus we have

$$\frac{d^j}{dx^j}\left[(x-x_0)^k\right] = \begin{cases} k(k-1)\dots(k-j+1)(x-x_0)^{k-j} & \text{if } j < k \\ k! & \text{if } j = k \\ 0 & \text{if } j > k \end{cases}$$

$$\frac{d^j}{dx^j}\left[(x-x_0)^k\right]\Big|_{x=x_0} = \begin{cases} 0 & \text{if } j < k \\ k! & \text{if } j = k \\ 0 & \text{if } j > k \end{cases}$$

Theorem 1.4.1. (Corollary of the General Mean Value Theorem)
Let $I \subseteq \mathbb{R}$ be an open interval, $x_0 \in I$, and $n \in \mathbb{N} \cup \{0\}$. Let $f : I \to \mathbb{R}$ have $n+1$ derivatives. Suppose $f^{(k)}(x_0) = 0 \quad \forall 0 \le k \le n$. Then for each point $x \neq x_0$ in the interval I, there exists a point c_{x,x_0} strictly between x and x_0 such that

$$f(x) = \frac{f^{(n+1)}(c_{x,x_0})}{(n+1)!}(x - x_0)^{n+1}$$

Proof. Here we will prove the claim for the case where $x > x_0$. The proof for $x < x_0$ is completely analogous. Let $g : I \to \mathbb{R}$ be defined by $g(t) = (t - x_0)^{n+1}$. Note that

$$g^{(k)}(x_0) = 0 \quad \forall 0 \le k \le n$$
$$g^{(n+1)}(t) = (n+1)! \quad \forall t \in I$$

Now, we apply the general mean value theorem to f and g on the interval $[x_0, x]$:

$$\begin{aligned} &\exists x_1 \in (x_0, x) \text{ such that } \frac{f'(x_1)}{g'(x_1)} = \frac{f(x) - f(x_0)}{g(x) - g(x_0)} \\ &\implies \frac{f'(x_1)}{g'(x_1)} = \frac{f(x)}{g(x)} \end{aligned} \tag{I}$$

Next, we apply the general mean value theorem to f' and g' on the interval $[x_0, x_1]$:

$$\begin{aligned} &\exists x_2 \in (x_0, x_1) \text{ such that } \frac{f''(x_2)}{g''(x_2)} = \frac{f'(x_1) - f'(x_0)}{g'(x_1) - g'(x_0)} \\ &\implies \frac{f''(x_2)}{g''(x_2)} = \frac{f'(x_1)}{g'(x_1)} \\ &\overset{(I)}{\implies} \frac{f''(x_2)}{g''(x_2)} = \frac{f(x)}{g(x)} \end{aligned}$$

Continuing in this way, we will obtain $x_{n+1} \in (x_0, x)$ such that

$$\frac{f^{(n+1)}(x_{n+1})}{g^{(n+1)}(x_{n+1})} = \frac{f(x)}{g(x)}$$

So,

$$\frac{f^{(n+1)}(x_{n+1})}{(n+1)!} = \frac{f(x)}{(x-x_0)^{n+1}}$$

Thus

$$f(x) = \frac{f^{(n+1)}(x_{n+1})}{(n+1)!}(x-x_0)^{n+1}$$

(We can use x_{n+1} as the c we were looking for) □

Question: What are the nicest functions that we know? Which functions are the easiest to work with?

Answer: Polynomials

General Question: Given a function f, is it possible to find a "good" approximation for f among polynomials?

Setup:

$*$) Let I be a nonempty open interval in $\mathbb{R}$
$*$) Let n be a nonnegative integer
$*$) Suppose $f : I \to \mathbb{R}$ has n derivatives and $x_0 \in I$
$*$) Suppose that we want to use the values

$$f(x_0), f'(x_0), ..., f^{(n)}(x_0)$$

to construct a polynomial approximation for f

What is the best we could hope for? Find a polynomial such that

$$\begin{aligned} p(x_0) &= f(x_0) \\ p'(x_0) &= f'(x_0) \\ &\vdots \\ p^{(n)}(x_0) &= f^{(n)}(x_0) \end{aligned}$$

Observation 2. Let x_0 be a fixed real number. A general polynomial of degree at most n can be expressed in powers of $(x - x_0)$ in the form

$$p(x) = c_0 + c_1(x-x_0) + ... + c_n(x-x_0)^n$$

Example 1.4.1. Consider $p(x) = x^2 - 3x - 1$. Let $x_0 = 1$. We can express $p(x)$ in powers of $x - 1$:

$$\begin{aligned} p(x) = x^2 - 3x - 1 &= [(x-1)+1]^2 - 3[(x-1)+1] - 1 \\ &= (x-1)^2 + 2(x-1) + 1 - 3(x-1) - 3 - 1 \\ &= (x-1)^2 - (x-1) - 3 \end{aligned}$$

Theorem 1.4.2. (Uniqueness of the Approximating Polynomial)
Let $I \subseteq \mathbb{R}$ be an open interval and $n \in \mathbb{N}$. Suppose $f : I \to \mathbb{R}$ has n derivatives ad $x_0 \in I$. Then there exists a unique polynomial $p(x)$ of degree at most n such that

$$\forall 0 \le l \le n \;\; p^{(l)}(x_0) = f^{(l)}(x_0), \text{ with } \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(x-x_0)^k$$

Proof. Let $p(x)$ be a general polynomial of degree at most n:

$$p(x) = c_0 + c_1(x-x_0) + ... + c_n(x-x_0)^n$$

Our goal is to show that

$$\text{If } \forall 0 \le l \le n \;\; p^{(l)}(x_0) = f^{(l)}(x_0) \text{ then } p(x) = \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k$$

Note that $p(x_0) = c_0$. Also, for $1 \le l \le n$ we have

$$\begin{aligned} p^{(l)}(x) &= \frac{d^l}{dx^l}\left[c_0 + \sum_{k=1}^{n} c_k(x - x_0)^k\right] \\ &= \frac{d^l}{dx^l}\left[\sum_{k=1}^{n} c_k(x - x)^k\right] \\ &= \sum_{k=1}^{n} c_k \frac{d^l}{dx^l}\left[(x - x_0)^k\right] \end{aligned}$$

Hence,

$$p^{(l)}(x_0) = \sum_{k=1}^{n} c_k \frac{d^l}{dx^l}\left[(x - x_0)^k\right]\Big|_{x=x_0} = c_l \cdot l!$$

Therefore,

$$\forall 1 \le l \le n \;\; p^{(l)}(x_0) = c_l \cdot l!$$

We conclude that

$$\begin{aligned} p \text{ agrees with } f \text{ to order } n \text{ at } x_0 &\iff \begin{cases} p(x_0) = f(x_0) \\ p^{(l)}(x_0) = f^{(l)}(x_0) \;\forall 1 \le l \le n \end{cases} \\ &\iff \begin{cases} c_0 = f(x_0) \\ l!c_l = f^{(l)}(x_0) \;\forall 1 \le l \le n \end{cases} \\ &\iff \begin{cases} c_0 = f(x_0) \\ c_l = \frac{f^{(l)}(x_0)}{l!} \;\forall 1 \le l \le n \end{cases} \\ &\iff p(x) = \sum_{k=0}^{n} c_k(x - x_0)^k \\ &= c_0 + \sum_{k=1}^{n} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k \\ &= f(x_0) + \sum_{k=1}^{n} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k \\ &= \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k \end{aligned}$$

□

Note . n^{th} Taylor Polynomial centered at 0 is called n^{th} Maclaurin Polynomial

Big Lesson: There is exactly one polynomial of degree at most n that satisfies

$$\begin{aligned} p(x_0) &= f(x_0) \\ p'(x_0) &= f'(x_0) \\ &\vdots \\ p^{(n)}(x_0) &= f^{(n)}(x_0) \end{aligned}$$

This polynomial is called the nth Taylor polynomial for f centered at x_0, and is given by

$$T_{n,x_0}(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2!}(x - x_0)^2 + \ldots + \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n$$

Theorem 1.4.3. (Taylor's Theorem with Lagarange Remainder)
Let $I \subseteq \mathbb{R}$ be an open interval, $x_0 \in I$, and $n \in \mathbb{N} \cup \{0\}$. Let $f : I \to \mathbb{R}$ have $n+1$ derivatives. Then for each point $x \neq x_0$ in I, there is a point c strictly between x and x_0 such that

$$f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{n!}(x - x_0)^k + \frac{f^{(n+1)}(c)}{(n+1)!}(x - x_0)^{n+1}$$

Remark 1.4.1. Note that clearly the above equality holds at $x = x_0$ too (for any value of c). Recall that for any fixed number R, $\lim_{n\to\infty} \frac{R^{n+1}}{(n+1)!} = 0$, however $f^{(n+1)}(c)$ may become very large.

Proof. Let $F_{n,x_0} = f(x) - T_{n,x_0}(x)$. Our goal is to show that

$$R_{n,x_0} = \frac{f^{(n+1)}(c)}{(n+1)!}(x - x_0)^{n+1}$$

for some c between x and x_0. Note that

$$\left.\begin{array}{l} f \text{ has } n+1 \text{ derivatives} \\ (i)\ T_{n,x+0} \text{ is a polynomial of degree } n, \text{ so it has } n+1 \text{ derivatives} \\ R_{n,x_0} = f - T \end{array}\right\} \implies R_{n,x_0} \text{ has } n+1 \text{ derivatives}$$

$(ii)\ \forall 0 \leq k \leq n \quad R^{(k)}_{n,x_0}(x_0) = f^{(k)}(x_0) - T^{(k)}_{n,x_0}(x_0) = 0$

$(i), (ii),$ Theorem 1.4.1 $\implies$ For each point $x \neq x_0$ in I, we have

$$R_{n,x_0}(x) = \frac{R^{(n+1)}_{n,x_0}(c)}{(n+1)!}(x - x_0)^{n+1} \text{ for some } c \text{ strictly between } x \text{ and } x_0 \qquad (I)$$

Now note that

$$R^{(n+1)}_{n,x_0}(c) = f^{(n+1)}(c) - T^{(n+1)}_{n,x_0}(c) = f^{(n+1)}(c) \qquad (II)$$

$$(I), (II) \implies R_{n,x_0} = \frac{f^{(n+1)}(c)}{(n+1)!}(x - x_0)^{(n+1)}$$

□

Chapter 2

Integration

2.1 The Riemann-Stieltjes Integral

Definition 2.1.1. (Almost Disjoint Intervals)
We say two intervals I and J are almost disjoint if either $I \cap J = \emptyset$ or $I \cap J$ has a single element.

Definition 2.1.2. (Partition)

First Viewpoint: A partition P of an interval $[a, b]$ is a finite set of points in $[a, b]$ that include both a and b. We always list the points of a partition $P = \{x_0, ..., x_n\}$ in increasing order:

$$x_0 = a < x_1 < x_2 < ... < b = x_n$$

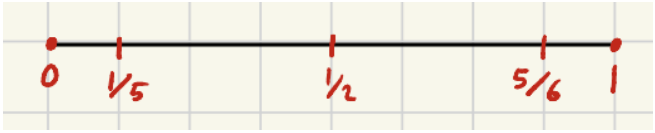
Second Viewpoint: A partition P of an interval $[a, b]$ is a finite collection of almost disjoint (nonempty) compact intervals whose union is $[a, b]$:

$$P = I_1, ..., I_n \text{ where } I_1 = [x_0, x_1], I_2 = [x_1, x_2], ..., I_n = [x_{n-1}, x_n] \quad (\text{with } x_0 = a, x_1 = b)$$

Example 2.1.1.

$$P = \left\{0, \frac{1}{5}, \frac{1}{2}, \frac{5}{6}, 1\right\}$$
$$P = \left\{[0, \frac{1}{5}], [\frac{1}{5}, \frac{1}{2}], [\frac{1}{2}, \frac{5}{6}], [\frac{5}{6}, 1]\right\}$$

are both partitions of $[0, 1]$.



Notation . Let $f : [a, b] \to \mathbb{R}$ be a bounded function. Let $P = \{x_0 = a, ..., x_n = b\}$ be a partition of $[a, b]$. We let

$$m = \inf\{f(x) : x \in [a, b]\}$$
$$M = \sup\{f(x) : x \in [a, b]\}$$
$$\forall 1 \leq k \leq n \quad m_k = \inf\{f(x) : x \in [x_{k-1}, x_k]\}$$
$$\forall 1 \leq k \leq n \quad M_k = \sup\{f(x) : x \in [x_{k-1}, x_k]\}$$

The existence of m, M, m_k, M_k as real numbers is guaranteed due to the assumption that f is bounded on $[a, b]$.

Remark 2.1.1. Suppose A and B are nonempty, bounded sets in $\mathbb{R}$ such that $A \subseteq B$. Then

(i) $\inf A \leq \sup A$

(ii) $\inf B \leq \sup B$

(iii) $\sup A \leq \sup B$

(iv) $\inf A \geq \inf B$

As a result

$$\forall 1 \leq k \leq n \quad m \leq m_k \leq M_k \leq M$$

Remark 2.1.2. If $f : [a,b] \to \mathbb{R}$ is continuous, then f attains its maximum and minimum over each compact subinterval. In this case,

$$m_k = \min\{f(x) : x \in [x_{k-1}, x_k]\}$$
$$M_k = \max\{f(x) : x \in [x_{k-1}, x_k]\}$$

Definition 2.1.3. (Lower Sum, Upper Sum)
Let $f : [a,b] \to \mathbb{R}$ be bounded and let $\alpha : [a,b] \to \mathbb{R}$ be increasing. Let $P = \{x_0, ..., x_n\}$ be a partition of $[a,b]$. Let $\Delta_{\alpha_k} = \alpha(x_k) - \alpha(x_{k-1})$.

(i) The lower Riemann-Stieltjes sum of f (R.S. sum of f) with respect to α for the partition P is defined by

$$L(f,\alpha,P) = \sum_{k=1}^{n} m_k\left(\alpha(x_k) - \alpha(x_{k-1})\right) = \sum_{k=1}^{n} m_k \Delta\alpha_k$$

(ii) The upper Riemann-Stieltjes sum of f (R.S sum of f) with respect α for the partition P is defined by

$$U(f,\alpha,P) = \sum_{k=1}^{n} M_k\left(\alpha(x_k) - \alpha(x_{k-1})\right) = \sum_{k=1}^{n} M_k \Delta\alpha_k$$

Note . Note that

$$m(\alpha(b) - \alpha(a)) \leq L(f,\alpha,P) \leq U(f,\alpha,P) \leq M(\alpha(b) - \alpha(a))$$

Indeed,

$$\begin{aligned} L(f,\alpha,P) = \sum_{k=1}^{n} m_k \Delta\alpha_k &\geq \sum_{k=1}^{n} m(\alpha(x_k) - \alpha(x_{k-1})) \\ &= m\sum_{k=1}^{n} \alpha(x_k) - \alpha(x_{k-1}) \\ &= m\left[\alpha(x_1) - \alpha(x_0) + \alpha(x_2) - \alpha(x_1) + ... + \alpha(x_n) - \alpha(x_{n-1})\right] \\ &= m\left[\alpha(x_n) - \alpha(x_0)\right] \\ &= m\left[\alpha(b) - \alpha(a)\right] \end{aligned}$$

Similarly,

$$\begin{aligned} U(f,\alpha,P) = \sum_{k=1}^{n} M_k \Delta\alpha_k &\leq \sum_{k=1}^{n} M(\alpha(x_k) - \alpha(x_{k-1})) \\ &= M\sum_{k=1}^{n} \alpha(x_k) - \alpha(x_{k-1}) \\ &= M\left[\alpha(x_1) - \alpha(x_0) + \alpha(x_2) - \alpha(x_1) + ... + \alpha(x_n) - \alpha(x_{n-1})\right] \\ &= M\left[\alpha(x_n) - \alpha(x_0)\right] \\ &= M\left[\alpha(b) - \alpha(a)\right] \end{aligned}$$

Notation . $\Pi[a,b]$, or Π for short, denotes the collection of all the possible partitions of $[a,b]$.

Definition 2.1.4. (Upper Riemann-Stieltjes Integral, Lower Riemann-Stieltjes Integral)
Let $f:[a,b]\to\mathbb{R}$ be bounded and $\alpha:[a,b]\to\mathbb{R}$ be increasing.

(*i*) The upper Riemann-Stieltjes integral of f with respect to α (on $[a,b]$) is defined by

$$U(f,\alpha)=\overline{\int_a^b} f\,d\alpha=\inf\{U(f,\alpha,P):P\in\Pi\}$$

(Note that the set $\{U(f,\alpha,P):P\in\Pi\}$ of all upper sums is bounded below by $m(\alpha(b)-\alpha(a))$, so the infimum is a real number)

(*ii*) The lower Riemann-Stieltjes integral of f with respect to α (on $[a,b]$) is defined by

$$L(f,\alpha)=\underline{\int_a^b} f\,d\alpha=\sup\{L(f,\alpha,P):P\in\Pi\}$$

Definition 2.1.5. (Riemann-Stieltjes Integrable Function)
Let $\alpha:[a,b]\to\mathbb{R}$ be an increasing function. A function $f:[a,b]\to\mathbb{R}$ is said to be Riemann-Stieltjes integrable (on $[a,b]$) with respect to α if

(*i*) f is bounded

(*ii*) $L(f,\alpha)=U(f,\alpha)$

In this case, the Riemann-Stieltjes integral of f with respect to α, denoted by

$$\int_a^b f\,d\alpha \text{ or } \int_a^b f(x)\,d\alpha \text{ or } \int_{[a,b]} f\,d\alpha$$

is the common value of $L(f,\alpha)$ and $U(f,\alpha)$. That is,

$$\int_a^b f\,d\alpha=L(f,\alpha)=U(f,\alpha)$$

Example 2.1.2. Let c be a fixed real number. Prove that the constant function $f(x)=c$ on $[a,b]$ is R.S. integrable and

$$\int_a^b f\,d\alpha=c(b-a)$$

Proof. For any partition $P=\{x_0,...,x_n\}$ of $[a,b]$ we have

$$\forall 1\le k\le n\quad m_k=\inf\{f(x):x\in[x_{k-1},x_k]\}=\inf\{c\}=c$$
$$M_k=\sup\{f(x):x\in[x_{k-1},x_k]\}=\sup\{c\}=c$$

Therefore,

$$\begin{aligned}L(f,\alpha,P)&=\sum_{k=1}^n m_k(\alpha(x_k)-\alpha(x_{k-1}))\\&=\sum_{k=1}^n c(\alpha(x_k)-\alpha(x_{k-1}))\\&=c\sum_{k=1}^n \alpha(x_k)-\alpha(x_{k-1})\\&=c[\alpha(b)-\alpha(a)]\end{aligned}$$

and

$$\begin{aligned} U(f,\alpha,P) &= \sum_{k=1}^{n} M_k(\alpha(x_k)-\alpha(x_{k-1})) \\ &= \sum_{k=1}^{n} c(\alpha(x_k)-\alpha(x_{k-1})) \\ &= c\sum_{k=1}^{n} \alpha(x_k)-\alpha(x_{k-1}) \\ &= c[\alpha(b)-\alpha(a)] \end{aligned}$$

Hence,

$$\begin{aligned} L(f,\alpha) &= \sup\{L(f,\alpha,P) : P\in\Pi\} = \sup\{c(\alpha(b)-\alpha(a))\} = c(\alpha(b)-\alpha(a)) \\ U(f,\alpha) &= \inf\{U(f,\alpha,P) : P\in\Pi\} = \inf\{c(\alpha(b)-\alpha(a))\} = c(\alpha(b)-\alpha(a)) \end{aligned}$$

Since $L(f,\alpha)=U(f,\alpha)=c(\alpha(b)-\alpha(a))$, we conclude that f is R.S. integrable with respect to α and

$$\int_a^b f d\alpha = c[\alpha(b)-\alpha(a)]$$

□

Example 2.1.3. Let $f:[a,b]\to\mathbb{R}$ be bounded and let $\alpha:[a,b]\to\mathbb{R}$ be a constant function ($\alpha(x)=c$). Prove that $\int_a^b f d\alpha = 0$.

Proof. For any partition $P=\{x_0,...,x_n\}$ of $[a,b]$,

$$\begin{aligned} L(f,\alpha,P) &= \sum_{k=1}^{n} m_k\left[\alpha(x_k)-\alpha(x_{k-1})\right] = \sum_{k=1}^{n} m_k\cdot 0 = 0 \\ U(f,\alpha,P) &= \sum_{k=1}^{n} M_k\left[\alpha(x_k)-\alpha(x_{k-1})\right] = \sum_{k=1}^{n} M_k\cdot 0 = 0 \end{aligned}$$

Therefore,

$$\begin{aligned} L(f,\alpha) &= \sup\{L(f,\alpha,P) : P\in\Pi\} = 0 \\ U(f,\alpha) &= \inf\{U(f,\alpha,P) : P\in\Pi\} = 0 \end{aligned}$$

So $L(f,\alpha)=U(f,\alpha)=0 \implies \int_a^b f d\alpha = 0$. □

Example 2.1.4. Let $f:[a,b]\to\mathbb{R}$ be defined by

$$f(x)=\begin{cases}1 & \text{if } x\in\mathbb{Q}\\ 0 & \text{if } x\notin\mathbb{Q}\end{cases}$$

Prove that

(*i*) if $\alpha:[a,b]\to\mathbb{R}$ is constant, then $\int_a^b f d\alpha = 0$ (proved in the last example)

(*ii*) if $\alpha:[a,b]\to\mathbb{R}$ is increasing and $\alpha(a)\neq\alpha(b)$, then $f\notin R_\alpha[a,b]$

Proof. (*ii*) For any partition $P=\{x_0,...,x_n\}$ of $[a,b]$ we have

$$\begin{aligned} \forall 1\le k\le n \quad m_k &= \inf\{f(x) : x\in[x_{k-1},x_k]\} = \inf\{0,1\} = 0 \\ M_k &= \sup\{f(x) : x\in[x_{k-1},x_k]\} = \sup\{0,1\} = 1 \end{aligned}$$

Therefore

$$L(f,\alpha,P)=\sum_{k=1}^{n} m_k\left[\alpha(x_k)-\alpha(x_{k-1})\right]=0\cdot[\alpha(b)-\alpha(a)]=0$$

$$U(f,\alpha,P)=\sum_{k=1}^{n} M_k\left[\alpha(x_k)-\alpha(x_{k-1})\right]=1\cdot[\alpha(b)-\alpha(a)]=\alpha(b)-\alpha(a)$$

Hence

$$\left.\begin{array}{l} L(f,\alpha)=\sup\{L(f,\alpha,P):P\in\Pi\}=0 \\ U(f,\alpha)=\inf\{U(f,\alpha,P):P\in\Pi\}=\alpha(b)-\alpha(a)\end{array}\right\}\implies L(f,\alpha)\neq U(f,\alpha)\implies f\notin R_\alpha[a,b]$$

□

Definition 2.1.6. (Refinement of a Partition)

First Viewpoint: A partition $Q=\{z_0,...,z_m\}$ of $[a,b]$ is a refinement of a partition $P=\{x_0,...,x_n\}$ of $[a,b]$ if $P\subseteq Q$. That is, if Q contains all points of P.

Second Viewpoint: A partition $Q=J_1,...,J_m$ of $[a,b]$ is a refinement of a partition $P=\{I_0,...,I_n\}$ of $[a,b]$ if every interval I_k of P is an almost disjoint union of one or more intervals of Q.

Example 2.1.5. Consider the following partitions of $[0,1]$:

$$P=\{0,\frac{1}{2},1\}$$
$$Q=\{0,\frac{1}{3},\frac{1}{2},1\}$$

Then Q is a refinement of P since $P\subseteq Q$.

Remark 2.1.3. Let P and Q be any two partitions of $[a,b]$. Then $P\cup Q$ will be a refinement of both P and Q because $P\subseteq P\cup Q$ and $Q\subseteq P\cup Q$. So, any two partitions of $[a,b]$ have a common refinement.

Example 2.1.6. Consider the following partitions of $[0,1]$:

$$P=\{0,\frac{1}{2},1\}$$
$$Q=\{0,\frac{1}{3},\frac{2}{3},1\}$$

P is not a refinement of Q and Q is not a refinement of P, but

$$P\cup Q=\{0,\frac{1}{3},\frac{1}{2},\frac{2}{3},1\}$$

is a refinement of both P and Q.

Definition 2.1.7. (Size of a Partition)
Let $P=\{x_0,...,x_n\}$ be a partition of $[a,b]$. The size of P, denoted $||P||$, is defined by

$$\begin{aligned}||P|| &= \max\{|x_k-x_{k-1}|:1\leq k\leq n\} \\ &= \max\{|x_1-x_0|,|x_2-x_1|,...,|x_n-x_{n-1}|\}\end{aligned}$$

2.2 9 Useful Theorems of Integrability

Theorem 2.2.1. (Inequalities of Refinements)
Let $f : [a, b] \to R$ be bounded and let $\alpha : [a, b] \to \mathbb{R}$ be increasing. Let P be a partition of $[a, b]$ and let Q be a refinement of P. Then

(i) $L(f, \alpha, P) \leq L(f, \alpha, Q)$

(ii) $U(f, \alpha, P) \geq U(f, \alpha, Q)$

Proof. Here, we will prove (i). The proof of (ii) is completely analogous. We proceed by induction on $l = \text{card}(Q \backslash P)$ (the number of points in $Q \backslash P$). Let $P = \{x_0, ..., x_n\}$.

Base Case: If $l = 0$, then

$$\left.\begin{array}{l} P \subseteq Q \\ \text{card}Q = \text{card}P \end{array}\right\} \implies P = Q \implies L(f, \alpha, P) = L(f, \alpha, Q)$$

If $l = 1$, then Q has exactly one extra point. Let's call this point z, so $\{z\} = Q \backslash P$. Note that

$$\left.\begin{array}{l} z \in [a, b] \\ P \text{ is a partition of } [a, b] \end{array}\right\} \implies \exists 1 \leq i \leq n \text{ such that } z \in (x_{i-1}, x_i)$$

Let

$$m_i' = \inf \{f(x) : x \in [x_{i-1}, z]\}$$
$$m_i'' = \inf \{f(x) : x \in [z, x_i]\}$$

Recall that if $A \subseteq B$, then $\inf A \geq \inf B$. Hence $m_i' \geq m_i$ and $m_i'' \geq m_i$ (where $m_i = \inf \{f(x) : x \in [x_{i-1}, x_i]\}$). We have

$$\begin{aligned} L(f, \alpha, P) &= \sum_{k=1}^{n} m_k \left[\alpha(x_k) - \alpha(x_{k-1})\right] \\ &= \left[\sum_{k=1, k\neq i}^{n} m_k \left(\alpha(x_k) - \alpha(x_{k-1})\right)\right] + m_i \left(\alpha(x_i) - \alpha(x_{i-1})\right) \\ &= \left[\sum_{k=1, k\neq i}^{n} m_k \left(\alpha(x_k) - \alpha(x_{k-1})\right)\right] + m_i \left(\alpha(z) - \alpha(x_{i-1})\right) + m_i \left(\alpha(x_i) - \alpha(z)\right) \\ &\leq \left[\sum_{k=1, k\neq i}^{n} m_k \left(\alpha(x_k) - \alpha(x_{k-1})\right)\right] + m_i' \left(\alpha(z) - \alpha(x_{i-1})\right) + m_i'' \left(\alpha(x_i) - \alpha(z)\right) \\ &= L(f, \alpha, Q) \end{aligned}$$

so

$$L(f, \alpha, P) \leq L(f, \alpha, Q)$$

Inductive Step: Suppose the claim is true for $l = r \geq 1$. We want to show the claim holds for $l = r + 1$. Suppose $\text{card}(Q \backslash P) = r + 1$. Let

$$Q \backslash P = \{z_1, ..., z_r, z_{r+1}\}$$

Let

$$\hat{Q} = P \cup \{z_1, ..., z_r\}$$

We have

$$L(f, \alpha, P) \overset{\text{hypoth.}}{\leq} L(f, \alpha, \hat{Q}) \overset{\text{base case}}{\leq} L(f, \alpha, Q)$$

So,

$$L(f, \alpha, P) \leq L(f, \alpha, Q)$$

□

Theorem 2.2.2. (Lower Sums are Smaller than Upper Sums)
Let $f : [a,b] \to \mathbb{R}$ be bounded and $\alpha : [a,b] \to \mathbb{R}$ be increasing. Let P_1 and P_2 be any two partitions of $[a,b]$. Then $L(f,\alpha,P_1) \leq U(f,\alpha,P_2)$.

Proof. Let $Q = P_1 \cup P_2$ be the common refinement of P_1 and P_2. We have

$$L(f,\alpha,P_1) \leq L(f,\alpha,Q) \leq U(f,\alpha,Q) \leq U(f,\alpha,P_2)$$

□

Theorem 2.2.3. (The Lower R.S. Integral is less than the Upper R.S. Integral)
Let $f : [a,b] \to \mathbb{R}$ be bounded and $\alpha : [a,b] \to \mathbb{R}$ be increasing. Then $L(f,\alpha) \leq U(f,\alpha)$.

Proof. Note that if A and B are nonempty subsets of $\mathbb{R}$ such that $\forall a \in A \quad \forall b \in B \quad a \leq b$, then $\sup A \leq \inf B$. Let $A = \{L(f,\alpha,P) : P \in \Pi\}$ and $B = \{U(f,\alpha,P) : P \in \Pi\}$. By Thm 2.2.2, $a \leq b$ for every $a \in A$ and $b \in B$. So, it follows that $\sup A \leq \sup B$, that is $L(f,\alpha) \leq U(f,\alpha)$. □

Theorem 2.2.4. (Cauchy Criterion for R.S. Integrability)
Let $f : [a,b] \to \mathbb{R}$ be bounded and let $\alpha : [a,b] \to \mathbb{R}$ be increasing.

$$f \in \mathbb{R}_\alpha[a,b] \iff \forall \epsilon > 0 \;\; \exists P_\epsilon \in \Pi \text{ such that } U(f,\alpha,P_\epsilon) - L(f,\alpha,P_\epsilon) < \epsilon.$$

Proof. ($\Longleftarrow$) Suppose $f : [a,b]$ is bounded and $\alpha : [a,b] \to \mathbb{R}$ is increasing, and

$$\forall \epsilon > 0 \; \exists P_\epsilon \in \Pi \text{ such that } U(f,\alpha,P_\epsilon) - L(f,\alpha,P_\epsilon) < \epsilon$$

We want to show $f \in R_\alpha[a,b]$. That is, we wat to show $L(f,\alpha) = U(f,\alpha)$. Since $U(f,\alpha) - L(f,\alpha) \geq 0$, it is enough to show that

$$\forall \epsilon > 0 \quad U(f,\alpha) - L(f,\alpha) < \epsilon$$

Let $\epsilon > 0$ be given. By assumption, there exists $P_\epsilon \in \Pi$ such that

$$U(f,\alpha,P_\epsilon) - L(f,\alpha,P_\epsilon) < \epsilon$$

We have

$$U(f,\alpha) = \inf\{U(f,\alpha,P) : P \in \Pi\} \leq U(f,\alpha,P_\epsilon)$$
$$L(f,\alpha) = \sup\{L(f,\alpha,P) : P \in \Pi\} \geq L(f,\alpha,P_\epsilon)$$

Hence,

$$L(f,\alpha,P_\epsilon) \leq L(f,\alpha) \overset{Thm 2.2.3}{\leq} U(f,\alpha) \leq U(f,\alpha,P_\epsilon)$$

The interval $[L(f,\alpha), U(f,\alpha)]$ is contained in the interval $[L(f,\alpha,P_\epsilon), U(f,\alpha,P_\epsilon)]$. Thus

$$U(f,\alpha) - L(f,\alpha) \leq U(f,\alpha,P_\epsilon) - L(f,\alpha,P_\epsilon) < \epsilon$$

as desired.

($\Longleftarrow$) Suppose $f \in R_\alpha[a,b]$. Our goal is to show

$$\forall \epsilon > 0 \quad \exists P_\epsilon \in \Pi \text{ such that } U(f,\alpha,P_\epsilon) - L(f,\alpha,P_\epsilon) < \epsilon$$

Let *epsilon* > 0 be given. Our goal is to find $P_\epsilon \in \Pi$ such that

$$U(f,\alpha,P_\epsilon) - L(f,\alpha,P_\epsilon) < \epsilon$$

Note that

$$U(f,\alpha) = \inf\{U(f,\alpha,P) : P \in \Pi\} \implies \exists P_1 \in \Pi \text{ such that } U(f,\alpha,P_1) < U(f,\alpha) + \frac{\epsilon}{2}$$
$$L(f,\alpha) = \sup\{L(f,\alpha,P) : P \in \Pi\} \implies \exists P_2 \in \Pi \text{ such that } L(f,\alpha) - \frac{\epsilon}{2} < L(f,\alpha,P_2)$$

Let $P_\epsilon = P_1 \cup P_2$. We claim that this partition can be used to prove the claim.

$$L(f,\alpha) - \frac{\epsilon}{2} < L(f,\alpha,P_2) \leq L(f,\alpha,P_\epsilon) \leq U(f,\alpha,P_\epsilon) \leq U(f,\alpha,P_1) < U(f,\alpha) + \frac{\epsilon}{2}$$

Thus

$$\begin{aligned} U(f,\alpha,P_\epsilon) - L(f,\alpha,P_\epsilon) &< [(U(f,\alpha)+\epsilon/2) - (L(f,\alpha)-\epsilon/2)] \\ &= U(f,\alpha) - L(f,\alpha) + \epsilon \\ &= 0 + \epsilon \\ &= \epsilon \end{aligned}$$

□

Theorem 2.2.5. (Refinement and Tag Stability for R.S. Integrals)
Suppose $f : [a,b] \to \mathbb{R}$ is bounded and $\alpha : [a,b] \to \mathbb{R}$ is increasing. Let $\epsilon > 0$ and let $P = \{x_0, ..., xn\}$ be a partition of $[a,b]$ such that $U(f\alpha, P) - L(f,\alpha,P) < \epsilon$. Then

(*i*) If Q is a refinement of P, then $U(f,\alpha,Q) - L(f,\alpha,Q) < \epsilon$.

(*ii*) If $\forall 1 \le k \le n$ t_k and s_k are arbitrary points in $[x_{k-1}, x_k]$, then

$$\sum_{k=1}^{n} |f(s_k) - f(t_k)| \, \Delta\alpha_k < \epsilon$$

(*iii*) If $f \in R_\alpha[a,b]$ and $\forall 1 \le k \le n$ $s_k \in [x_{k-1}, x_k]$, then

$$\left| \sum_{k=1}^{n} f(s_k)\Delta\alpha_k - \int_a^b f d\alpha \right| < \epsilon$$

Proof. (*i*) We have

$$L(f,\alpha,P) \le L(f,\alpha,Q) \le U(f,\alpha,Q) \le U(f,\alpha,P)$$

Therefore,

$$U(f,\alpha,Q) - L(f,\alpha,Q) \le U(f,\alpha,P) - L(f,\alpha,P) < \epsilon$$

(*ii*) For each $1 \le k \le n$, we have

$$\begin{aligned} &m_k \le f(s_k) \le M_k \\ &m_k \le f(t_k) \le M_k \iff -M_k \le -f(t_k) \le -m_k \end{aligned}$$

so

$$\begin{aligned} &m_k - M_k \le f(s_k) - f(t_k) \le M_k - m_k \\ \implies &-(M_k - m_k) \le f(s_k) - f(t_k) \le M_k - m_k \\ \implies &|f(s_k) - f(t_k)| \le M_k - m_k \end{aligned}$$

Hence,

$$\sum_{k=1}^{n} |f(s_k) - f(t_k)| \, \Delta\alpha_k \le \sum_{k=1}^{n} (M_k - m_k) \, \Delta\alpha_k = U(f,\alpha,P) - L(f,\alpha,P) < \epsilon$$

(*iii*) $\forall 1 \le k \le n$ $m_k \le f(s_k) \le M_k$ so,

$$\sum_{k=1}^{n} m_k \Delta\alpha_k \le \sum_{k=1}^{n} f(s_k)\Delta\alpha_k \le \sum_{k=1}^{n} M_k \Delta\alpha_k$$

Therefore,

$$L(f,\alpha,P) \le \sum_{k=1}^{n} f(s_k)\Delta\alpha_k \le U(f,\alpha,P)$$

Also, note that

$$L(f,\alpha,P) \le \int_a^b f d\alpha \le U(f,\alpha,P)$$

Hence,

$$\left| \sum_{k=1}^{n} f(s_k)\Delta\alpha_k - \int_a^b f d\alpha \right| \le U(f,\alpha,P) - L(f,\alpha,P) < \epsilon$$

□

Remark 2.2.1. Let $f : [a,b] \to \mathbb{R}$ be bounded, $\alpha : [a,b] \to \mathbb{R}$ be increasing, and $P = \{x_0, ..., x_n\}$ be a partition of $[a,b]$.

* If for every $1 \leq k \leq n$ we choose a point $t_k \in [x_{k-1}, x_k]$, then we call the pair $(P, (t_k)_{1\leq k\leq n})$ a tagged partition.
* We sometimes refer to $(t_k)_{1\leq k\leq n}$ as a tag of a partition.
* A sum of the form
$$S\left(f, \alpha, (P, (t_k)_{1\leq k\leq n})\right) = \sum_{k=1}^{n} f(t_k)\Delta\alpha_k$$
is called a Riemann-Stieltjes sum of f with respect to α (The Riemann-Stieltjes sum of f with respect to α for the tagged partition $(P, (t_k)_{1\leq k\leq n})$).
* As a direct consequence of Theorem 2.2.5 and the Cauchy Criterion for Riemann-Stieltjes integrability, we have
$$f \in R_\alpha[a,b]$$
$$\implies \forall \epsilon > 0 \;\; \exists P_\epsilon \in \Pi \text{ such that } \forall P_\epsilon \subseteq P \;\; \forall \text{tag}(t_k) \text{ of } P \;\; \left| S(f, \alpha, (P, (t_k)_{1\leq k\leq n})) - \int_a^b f d\alpha \right| < \epsilon$$
The converse is true in the following sense:
Suppose $f : [a,b] \to \mathbb{R}$ is bounded and $\alpha : [a,b] \to \mathbb{R}$ is increasing. Then the following two statements are equivalent:

(*i*) $f \in R_\alpha[a,b]$

(*ii*) There exists a number A such that
$$\forall \epsilon > 0 \;\; \exists P_\epsilon \in \Pi \text{ such that } \forall P_\epsilon \subseteq P \;\; \forall \text{tag}(t_k) \text{ of } P \;\; |S(f, \alpha, (P, (t_k)_{1\leq k\leq n})) - A| < \epsilon$$

Lemma 2.2.1. Let $f : [a,b] \to \mathbb{R}$ be a bounded function. Let $P = \{x_0, ..., x_n\}$ be a partition of $[a,b]$. Then
$$\forall 1 \leq k \leq n \quad \sup\{|f(s) - f(t)| : s, t \in [x_{k-1}, x_k]\} = M_k - m_k$$

Proof. Let $k \in \{1, ..., n\}$. We need to show

(*i*) $\forall s, t \in [x_{k-1}, x_k] \; |f(s) - f(t)| \leq M_k - m_k$

(*ii*) $\forall \epsilon > 0 \;\; \exists \hat{s}, \hat{t} \in [x_{k-1}, x_k]$ such that $M_k - m_k < \left|f(\hat{s}) - f(\hat{t})\right|$

We proved (*i*) in our discussion of theorem 2.2.5. It remains to show (*ii*):
Let $\epsilon > 0$ be given. Then
$$m_k = \inf\{f(t) : [x_{k-1}, x_k]\} \implies \exists \hat{t} \in [x_{k-1}, x_k] \text{ such that } f(\hat{t}) < m_k + \frac{\epsilon}{2} \implies -m_k - \frac{\epsilon}{2} < -f(\hat{t})$$
$$M_k = \sup\{f(t) : [x_{k-1}, x_k]\} \implies \exists \hat{s} \in [x_{k-1}, x_k] \text{ such that } M_k - \frac{\epsilon}{2} < f(\hat{s})$$
Therefore, adding the inequalities gives us
$$M_k - m_k - \epsilon < f(\hat{s}) - f(\hat{t}) \leq \left|f(\hat{s}) - f(\hat{t})\right|$$
□

Theorem 2.2.6. (Continuity Implies Integrability)
Let $f : [a,b] \to \mathbb{R}$ be continuous and let $\alpha : [a,b] \to \mathbb{R}$ be increasing. Then $f \in R_\alpha[a,b]$.

Proof. First, note that
$$\left.\begin{array}{l} f : [a,b] \to \mathbb{R} \text{ is continuous} \\ [a,b] \text{ is compact} \end{array}\right\} \overset{\text{EVT}}{\implies} f \text{ is bounded on } [a,b]$$

Now, according to the Cauchy Criterion of R.S. integrability, it is enough to show that

$$\forall \epsilon > 0 \quad \exists P \in \Pi \text{ such that } U(f,\alpha,P) - L(f,\alpha,P) < \epsilon$$

Let $\epsilon > 0$ be given. Our goal is to find $P \in \Pi$ such that

$$U(f,\alpha,P) - L(f,\alpha,P) < \epsilon \qquad (*)$$

We have

$$\left.\begin{array}{l} f \text{ is continuous on } [a,b] \\ [a,b] \text{ is compact} \end{array}\right\} \implies f \text{ is uniformly continuous on } [a,b]$$

So, for the given *epsilon*,

$$\exists \delta \text{ such that } \forall s,t \in [a,b] \quad |s-t| < \delta \implies |f(s)-f(t)| < \frac{\epsilon}{2[\alpha(b)-\alpha(a)+1]}$$

Let P be any partition of $[a,b]$ such that $\|P\| < \delta$. We claim that $(*)$ holds for such a P. Indeed, $\forall k \in \{1,...,n\} \quad \forall s,t \in [x_{k-1},x_k]$ we have

$$|s-t| < \delta \implies |f(t)-f(s)| < \frac{\epsilon}{2[\alpha(b)-\alpha(a)+1]}$$

Thus

$$\sup_{s,t\in[x_{k-1},x_k]} |f(s)-f(t)| \leq \frac{\epsilon}{2[\alpha(b)-\alpha(a)+1]}$$

Hence, by lemma 2.2.1,

$$M_k - m_k \leq \frac{\epsilon}{2[\alpha(b)-\alpha(a)+1]}$$

Therefore,

$$\begin{aligned} U(f,\alpha,P) - L(f,\alpha,P) &= \sum_{k=1}^{n} M_k - m_k \Delta\alpha_k \\ &\leq \sum_{k=1}^{n} \frac{\epsilon}{2[\alpha(b)-\alpha(a)+1]} \Delta\alpha_k \\ &= \frac{\epsilon}{2[\alpha(b)-\alpha(a)+1]} [\alpha(b)-\alpha(a)] \\ &\leq \frac{\epsilon}{2} \\ &< \epsilon \end{aligned}$$

□

Example 2.2.1. $f : [0,10] \to \mathbb{R}$ defined by $f(x) = \begin{cases} \frac{sinx}{x} & x > 0 \\ 1 & x = 0 \end{cases}$ is continuous.
$\alpha : [0,10] \to \mathbb{R}$ defined by $\alpha(x) = \lfloor x \rfloor$ is an increasing function.
Hence, by theorem 2.2.6,

$$\int_0^{10} f(x) d\lfloor x \rfloor \text{ exists}$$

Lemma 2.2.2. Let $\alpha : [a,b] \to \mathbb{R}$ be increasing and continuous and let $\alpha(a) < \alpha(b)$. Then for each $n \in \mathbb{N}$, there exists a partition $P = \{x_0,...,x_n\}$ such that

$$\forall 1 \leq k \leq n \quad \Delta\alpha_k = \alpha(x_k) - \alpha(x_{k-1}) = \frac{\alpha(b)-\alpha(a)}{n}$$

Proof. Let $n \in \mathbb{N}$. Divide the interval $[\alpha(a),\alpha(b)]$ into n subintervals of equal length $\frac{\alpha(b)-\alpha(a)}{n}$. For each $1 < k < n$, we have $y_k \in (\alpha(a),\alpha(b))$, hence by the intermediate value theorem

$$\exists x_k \in (a,b) \text{ such that } y_k = \alpha(x_k)$$

Since α is increasing, we have $a < x_1 < x_2 < ... < x_{n-1} < b$. Now, if we define $x_0 = a$ and $x_n = b$, then $P = \{x_0, ..., x_n\}$ will be a partition of $[a, b]$ such that

$$\forall 1 \leq k \leq n \quad \delta\alpha_k = \alpha(x_k) - \alpha(x_{k-1}) = y_k - y_{k-1} = \frac{\alpha(b) - \alpha(a)}{n}$$

□

Theorem 2.2.7. (Monotone Functions are Riemann-Stieltjes Integrable)
Suppose $\alpha : [a, b] \to \mathbb{R}$ is increasing and continuous.

(i) If $f : [a, b] \to \mathbb{R}$ is increasing, then $f \in R_\alpha[a, b]$

(ii) If $f : [a, b] \to \mathbb{R}$ is decreasing, then $f \in R_\alpha[a, b]$

Proof. We prove (i) ((ii) is analogous). First note that

$$\forall x \in [a, b] \quad f(a) \leq f(x) \leq f(b) \implies f \text{ is bounded on } [a, b]$$

If $\alpha(a) = \alpha(b)$, then we previously proved that $f \in R_\alpha[a, b]$ and $\int_a^b f d\alpha = 0$. So it remains to prove the claim for the case where $\alpha(a) \neq \alpha(b)$. According to the Cauchy Criterion for integrability, in order to show $f \in R_\alpha[a, b]$, it is enough to show that

$$\forall \epsilon > 0 \quad \exists P \in \Pi \text{ such that } U(f, \alpha, P) - L(f, \alpha, P) < \epsilon.$$

Let $\epsilon > 0$ be given. Our goal is to find $P \in Pi$ such that

$$U(f, \alpha, P) - L(f, \alpha, P) < \epsilon$$

Choose $n \in N$ large enough so that

$$\frac{\alpha(b) - \alpha(a)}{n}[f(b) - f(a)] < \epsilon.$$

Let $\tilde{P} = \{x_0, ..., x_n\}$ be a partition of $[a, b]$ such that

$$\begin{aligned} \forall 1 \leq k \leq n \quad \delta\alpha_k &= \alpha(x_k) - \alpha(x_{k-1}) \\ &= \frac{\alpha(b) - \alpha(a)}{n} \end{aligned}$$

We claim that $\tilde{P}$ can be used as the P we are looking for. Indeed,

$$\begin{aligned} U(f, \alpha, \tilde{P}) - L(f, \alpha, \tilde{P}) &= \sum_{k=1}^{n} (M_k - m_k)\delta\alpha_k \\ &= \frac{\alpha(b) - \alpha(a)}{n} \sum_{k=1}^{n} (M_k - m_k) \\ &= \frac{\alpha(b) - \alpha(a)}{n} \sum_{k=1}^{n} [f(x_k) - f(x_{k-1})] \quad \text{(for all } k \ M_k = \sup f(x) = f(x_k) \text{ and } m_k = f(x_{k-1})) \\ &= \frac{\alpha(b) - \alpha(a)}{n}[f(b) - f(a)] \\ &< \epsilon \end{aligned}$$

as desired. □

Example 2.2.2. $\alpha : [0, 2] \to \mathbb{R}$ given by $\alpha(x) = x^2$ is continuous and increasing and $f : [0, 2] \to \mathbb{R}$ given by $f(x) = \lfloor x^2 \rfloor$ is increasing. By theorem 2.2.7, $\int_0^2 \lfloor x^2 \rfloor d(x^2)$ exists.

Remark 2.2.2. Recall that to show $f \in R_\alpha[a, b]$, it is enough to show

$$\forall \epsilon > 0 \quad \exists P \in \Pi \text{ such that } U(f, \alpha, P) - L(f, \alpha, P) < \epsilon,$$

Also, note that

$$U(f,\alpha,P)-L(f,\alpha,P)=\sum_{k=1}^{n}(M_k-m_k)\delta\alpha_k.$$

So, we would like to make $\sum(M_k-m_k)\delta\alpha_k$ small.

$*$) Sometimes, this can be done by ensuring that M_k-m_k is small

$$M_k-m_k<\epsilon \implies \sum(M_k-m_k)\delta\alpha_k<\epsilon\sum\delta\alpha_k=\epsilon[\alpha(b)-\alpha(a)]$$

$*$) Sometimes, this can be done by making $\delta\alpha_k$ small

$$\sum(M_k-m_k)\delta\alpha_k \overset{\text{Lemma2.2.1}}{\leq} \sum 2\tilde{M}\delta\alpha_k=2\tilde{M}\sum\delta\alpha_k$$

where $\tilde{M}=\sup|f(x)|$.

Theorem 2.2.8. (Finite Discontinuities Implies R.S. Integrability)
Let $f:[a,b]\to\mathbb{R}$ be bounded and suppose f has finitely many discontinuities: $y_1<y_2<...<y_n$. Suppose $\alpha:[a,b]\to\mathbb{R}$ is increasing and α is continuous at $y_1<y_2<...<y_n$. Then $f\in R_\alpha[a,b]$.

Proof. According to the Cauchy criterion for Riemann-Stieltjes integrability, it is enough to show that

$$\forall\epsilon>0 \ \ \exists P\in\Pi \text{ such that } U(f,\alpha,P)-L(f,\alpha,P)<\epsilon$$

Let $\epsilon>0$ be given. Our goal is to find P such that

$$U(f,\alpha,P)-L(f,\alpha,P)<\epsilon$$

Let $\tilde{M}=\sup\limits_{a\leq x\leq b}|f(x)|$. Let $\hat{\epsilon}=\frac{\epsilon}{[\alpha(b)-\alpha(a)]+2\tilde{M}+1}$. We make two claims:

Claim 1: There exists finitely many disjoint intervals $[u_!,v_1],...,[u_m,v_m]$ such that

(i) $\forall 1\leq j\leq m \ \ y_j\in[u_j,v_j]$.
(ii) $\forall 1\leq j\leq m$ if $y_j\notin\{a,b\}$ then $y_j\in(u_j,v_j)$.
(iii) $\forall 1\leq j\leq m \ \ \alpha(v_j)-\alpha(u_j)<\frac{\hat{\epsilon}}{m}$ and so $\sum_{j=1}^{m}\alpha(v_j)-\alpha(u_j)<\epsilon$.

Claim 2: Let $K=[a,b]\setminus\bigcup_{j=1}^{m}(u_j,v_j)$. Then f is uniformly continuous on K.

We will prove claim 1 and claim 2 shortly, for now assume they hold. By claim 2,

$$\exists\delta>0 \text{ such that } \forall s,t\in K \text{ if } |s-t|<\delta \text{ then } |f(s)-f(t)|<\hat{\epsilon}.$$

Now, we form a partition $\tilde{P}$ of $[a,b]$ as follows:

(i) $\forall 1\leq j\leq m \ \ u_j,v_j\in\tilde{P}$.

(ii) $\forall 1\leq j\leq m$ no point of the segment (u_j,v_j) is in $\tilde{P}$.

(iii) If $1\leq k\leq n$ is such that $x_{k-1}\notin\{u_1,...,u_m\}$, then we choose x_k such that $x_k-x_{k-1}<\delta$.

We claim that $\tilde{P}$ can be used as the P that we were looking for. Indeed, let

$$A=\{k:x_{k-1}\notin\{u_1,...,u_m\}\},B=\{1,...,n\}\setminus A.$$

Note that

$*$) if $k\in A$, then $x_k-x_{k-1}<\delta$, so

$$\begin{aligned}
&\forall s,t\in[x_{k-1},x_k] \ \ |s-t|<\epsilon\\
\implies &\forall s,t\in[x_{k-1},x_k] \ \ |f(s)-f(t)|<\hat{\epsilon}\\
\implies &\sup_{s,t\in[x_{k-1},x_k]}|f(s)-f(t)|\leq\hat{\epsilon}\\
\implies &M_k-m_k\leq\hat{\epsilon}
\end{aligned}$$

$*$) if $k \in B$, then

$$M_k - m_k = \sup_{s,t \in [x_{k-1}, x_k]} |f(s) - f(t)| \leq 2\tilde{M}$$

Therefore,

$$\begin{aligned} U(f, \alpha, \tilde{P}) - L(f, \alpha, \tilde{P}) &= \sum_{k=1}^{n} (M_k - m_k)\Delta\alpha_k \\ &= \sum_{k \in A} (M_k - m_k)\Delta\alpha_k + \sum_{k \in B} (M_k - m_k)\Delta\alpha_k \\ &\leq \sum_{k \in A} \hat{\epsilon}\Delta\alpha_k + \sum_{k \in B} 2\tilde{M}\Delta\alpha_k \\ &\leq \tilde{\epsilon} \sum_{k=1}^{n} \Delta\alpha_k + 2\tilde{M} \sum_{k \in B} \Delta\alpha_k \\ &\leq \tilde{\epsilon}[\alpha(b) - \alpha(a)] + 2\tilde{M}\hat{\epsilon} \\ &= [\alpha(b) - \alpha(a) + 2\tilde{M}]\hat{\epsilon} \\ &< \epsilon \end{aligned}$$

Proof of Claim 1: Since for each $q \leq j \leq m$ α is continuous at y_j, we can choose $\delta_j > 0$ such that

$$\text{if } |y - y_j| < \delta_j \text{ then } |\alpha(y) - \alpha(y_j)| < \frac{\hat{\epsilon}}{2M}.$$

Now let

$$\tilde{\delta} = \frac{\min}{4} \{\delta_1, \delta_2, ..., \delta_m, y_2 - y_1, y_3 - y_2, y_4 - y_3, ..., y_m - y_{m-1}\}.$$

For each $1 \leq j \leq m$, we define

(i) If $y_j \notin \{a, b\}$, then $[u_j, v_j] = [y_j - \tilde{\delta}, y_j + \tilde{\delta}]$.

(ii) If $y_j = a$, then $[u_j, v_j] = [a, a + \tilde{\delta}]$.

(iii) If $y_j = b$, then $[u_j, v_j] = [b - \tilde{\delta}, b]$.

Clearly, these intervals satisfy all the requirements. In particular,

$$\alpha(v_j) - \alpha(u_j) = |\alpha(v_j) - \alpha(u_j)| \leq |\alpha(v_j) - \alpha(y_j)| + |\alpha(y_j) - \alpha(u_j)| < \frac{\hat{\epsilon}}{2M} + \frac{\hat{\epsilon}}{2M} = \frac{\hat{\epsilon}}{M}$$

Proof of Claim 2: Note that $\bigcup_{j=1}^{m} (u_j, v_j)$ is open. Hence,

$$K = [a, b] \backslash \bigcup_{j=1}^{m} (u_j, v_j) = [a, b] \cap [\bigcup_{j=1}^{m} (u_j, v_j)]^c \implies K \text{ is closed.}$$

$$\left.\begin{array}{l} K \subseteq [a, b] \\ K \text{ is closed} \\ [a, b] \text{ is compact} \end{array}\right\} \implies K \text{ is compact.}$$

Since $f : K \to \mathbb{R}$ is continuous and K is compact, we can conclude that f is uniformly continuous on K.

□

Theorem 2.2.9. (Applying Continuous Functions Preserves Integrability)
Suppose $f \in R_\alpha[a, b]$ such that $\forall x \in [a, b] \;\; m \leq f(x) \leq M$ and suppose $\varphi : [m, M] \to \mathbb{R}$ is continuous. Then $\varphi \circ f \in R_\alpha[a, b]$.

Proof. First, note that a composition of bounded functions is bounded, so $h = \varphi \circ f$ is bounded. According to the Cauchy criterion, in order to show $h \in R_\alpha[a, b]$ it is enough to show that

$$\forall \epsilon > 0 \; \exists P \in \Pi \text{ such that } U(h, \alpha, P) - L(h, \alpha, P) < \epsilon.$$

Let $\epsilon > 0$ be given. Our goal is to find $P \in \Pi$ such that

$$U(h, \alpha, P) - L(h, \alpha, P) < \epsilon.$$

Let $\tilde{M} = \sup_{a \le x \le b} |h(x)|$. Let $\hat{\epsilon} = \frac{\epsilon}{[\alpha(b)-\alpha(a)]+2\tilde{M}+1}$. We have

$$\left.\begin{array}{l} \varphi \text{ is continuous on } [m, M] \\ [m, M] \text{ is compact} \end{array}\right\} \implies \varphi \text{ is uniformly continuous on } [m, M].$$

so

$$\exists 0 < \delta < \hat{\epsilon} \text{ such that } \forall s, t \in [m, M] \text{ if } |s - t| < \delta \text{ then } |\varphi(s) - \varphi(t)| < \hat{\epsilon}$$

Also,

$$f \in R_\alpha[a, b] \implies \exists \tilde{P} \text{ such that } U(f, \alpha, \tilde{P}) - L(f, \alpha, \tilde{P}) = \sum_{k=1}^{n} (M_k - m_k)\, \Delta\alpha_k < \delta^2$$

We claim that this $\tilde{P}$ can be used as the P we are looking for. Indeed, let

$$\forall 1 \le k \le n \quad m_k^* = \inf \{h(x) : x \in [x_{k-1}, x_k]\}, \quad M_k^* = \sup \{h(x) : x \in [x_{k-1}, x_k]\}.$$

Note that

$$U(h, \alpha, \tilde{P}) - L(h, \alpha, \tilde{P}) = \sum_{k=1}^{n} (M_k^* - m_k^*)\, \Delta\alpha_k$$

In what follows, we will show that the above sum is less than ϵ. Divide the numbers $1, ..., n$ into two cases:

$$A = \{k : M_k - m_k < \delta\} \quad B = \{k : M_k - m_k \ge \delta\}$$

We have

$$U(h, \alpha, \tilde{P}) - L(h, \alpha, \tilde{P}) = \sum_{k=1}^{n} (M_k^* - m_k^*)\, \Delta\alpha_k = \sum_{k \in A} (M_k^* - m_k^*)\, \Delta\alpha_k + \sum_{k \in B} (M_k^* - m_k^*)\, \Delta\alpha_k \qquad \text{(I)}$$

$*$) If $k \in A$:

$$\begin{aligned} M_k - m_k < \delta &\implies \sup_{x \in [x_{k-1}, x_k]} |f(x) - f(y)| < \delta \\ &\implies \forall x, y \in [x_{k-1}, x_k] \quad |f(x) - f(y)| < \delta \\ &\implies \forall x, y \in [x_{k-1}, x_k] \quad |\varphi(f(x)) - \varphi(f(y))| < \hat{\epsilon} \\ &\implies \forall x, y \in [x_{k-1}, x_k] \quad |h(x) - h(y)| < \hat{\epsilon} \\ &\implies \sup_{x, y \in [x_{k-1}, x_k]} |h(x) - h(y)| \le \hat{\epsilon} \\ &\implies M_k^* - m_k^* \le \hat{\epsilon} \end{aligned} \qquad \text{(II)}$$

$*$) If $k \in B$, then

$$\delta \sum_{k \in B} \Delta\alpha_k = \sum_{k \in B} \delta \Delta\alpha_k \le \sum_{k \in B} (M_k - m_k)\, \Delta\alpha_k \le \sum_{k=1}^{n} (M_k - m_k)\, \Delta\alpha_k = U(f, \alpha, \tilde{P}) - L(f, \alpha, \tilde{P}) < \delta^2$$

$$\implies \delta \sum_{k \in B} \Delta\alpha_k < \delta^2 \implies \sum_{k \in B} \Delta\alpha_k < \delta \implies \sum_{k \in B} \Delta\alpha_k < \hat{\epsilon} \qquad \text{(III)}$$

It folows from (I), (II), and (III) that

$$\begin{aligned}
\sum_{k=1}^{n}(M_k^*-m_k^*)\,\Delta\alpha_k &= \sum_{k\in A}(M_k^*-m_k^*)\,\Delta\alpha_k+\sum_{k\in B}(M_k^*-m_k^*)\,\Delta\alpha_k\\
&\le \sum_{k\in A}\hat{\epsilon}\Delta\alpha_k+\sum_{k\in B}2\tilde{M}\Delta\alpha_k\\
&= \hat{\epsilon}\sum_{k\in A}\Delta\alpha_k+2\tilde{M}\sum_{k\in B}\Delta\alpha_k\\
&\le \hat{\epsilon}\sum_{k=1}^{n}\Delta\alpha_k+2\tilde{M}\hat{\epsilon}\\
&= \hat{\epsilon}[\alpha(b)-\alpha(a)+2\tilde{M}]\\
&= \frac{\epsilon}{\alpha(b)-\alpha(a)+2\tilde{M}+1}[\alpha(b)-\alpha(a)+2\tilde{M}]\\
&< \epsilon
\end{aligned}$$

□

Example 2.2.3. If $f\in R_\alpha[a,b]$, then $|f|\in R_\alpha[a,b]$.
If $f\in R_\alpha[a,b]$, then $f^2\in R_\alpha[a,b]$.

2.3 The Sequential Criterion of Integrability

Theorem 2.3.1. (The Sequential Criterion of Integrability)
Suppose $f : [a, b] \to \mathbb{R}$ is bounded and $\alpha : [a, b] \to \mathbb{R}$ is increasing.

(i) If $f \in R_\alpha[a, b]$, then there exists a sequence of partitions $(P_n)_{n\geq 1}$ in $\Pi[a, b]$ such that

$$\lim_{n\to\infty} [U(f, \alpha, P_n) - L(f, \alpha, P_n)] = 0.$$

(ii) If there exists a sequence of partitions $(P_n)_{n\geq 1}$ in $\Pi[a, b]$ such that

$$\lim_{n\to\infty} [U(f, \alpha, P_n) - L(f, \alpha, P_n)] = 0,$$

then $f \in R_\alpha[a, b]$ and

$$\int_a^b f d\alpha = \lim_{n\to\infty} U(f, \alpha, P_n) = \lim_{n\to\infty} L(f, \alpha, P_n).$$

Proof. (i) $f \in R_\alpha[a, b] \iff \forall \epsilon > 0, \quad \exists P_\epsilon \in \Pi$ such that $U(f, \alpha, P_\epsilon) - L(f, \alpha, P_\epsilon) < \epsilon$. In particular,

$$\begin{aligned}
\epsilon = 1 &\implies \exists P_1 \in \Pi \text{ such that } U(f, \alpha, P_1) - L(f, \alpha, P_1) < 1 \\
\epsilon = \frac{1}{2} &\implies \exists P_2 \in \Pi \text{ such that } U(f, \alpha, P_2) - L(f, \alpha, P_2) < \frac{1}{2} \\
\epsilon = \frac{1}{3} &\implies \exists P_3 \in \Pi \text{ such that } U(f, \alpha, P_3) - L(f, \alpha, P_3) < \frac{1}{3} \\
&\vdots
\end{aligned}$$

In this way, we obtain a sequence $(P_n)_{n\geq 1}$ of partitions such that

$$\forall n \geq 1 \quad U(f, \alpha, P_n) - L(f, \alpha, P_n) < \frac{1}{n}.$$

It follows from the squeeze theorem that

$$\lim_{n\to\infty} [U(f, \alpha, P_n) - L(f, \alpha, P_n)] = 0.$$

(ii) According the Cauchy Criterion, it is enough to show that

$$\forall \epsilon > 0 \quad \exists P \in \Pi[a, b] \text{ such that } U(f, \alpha, P_n) - L(f, \alpha, P_n) < \epsilon.$$

Let $\epsilon > 0$ be given. Our goal is to find $P \in \Pi$ such that

$$U(f, \alpha, P) - L(f, \alpha, P) < \epsilon.$$

Since

$$\lim_{n\to\infty} [U(f, \alpha, P_n) - L(f, \alpha, P_n)] = 0,$$

there exists $N \in \mathbb{N}$ such that

$$\forall n \in \mathbb{N} \quad U(f, \alpha, P_n) - L(f, \alpha, P_n) < \epsilon.$$

In particular, P_{N+1} can be used as the P we were looking for. It remains to show that $\int_a^b f d\alpha = \lim_{n\to\infty} U(f, \alpha, P_n) = \lim_{n\to\infty} L(f, \alpha, P_n)$. We have

$$0 \leq U(f, \alpha, P_n) - U(f, \alpha) \leq U(f, \alpha, P_n) - L(f, \alpha, P_n) \leq U(f, \alpha, P_n) - L(f, \alpha, P_n) \to 0 \text{ as } n \to \infty.$$

If follows from the squeeze theorem that

$$\lim_{n\to\infty} [U(f, \alpha, P_n) - U(f, \alpha)] = 0.$$

Thus

$$\lim_{n\to\infty} U(f, \alpha, P_n) = U(f, \alpha) = \int_a^b f d\alpha.$$

Similary,

$$0 \leq L(f,\alpha) - L(f,\alpha,P_n) \leq U(f,\alpha) - L(f,\alpha,P_n) \leq U(f,\alpha,P_n) - L(f,\alpha,P_n) \to 0 \text{ as } n \to \infty$$
$$\overset{\text{SQZ Thm}}{\Longrightarrow} \lim_{n\to\infty} [L(f,\alpha) - L(f,\alpha,P_n)]$$
$$\Longrightarrow \lim_{n\to\infty} L(f,\alpha,P_n) = L(f,\alpha) = \int_a^b f d\alpha.$$

□

Theorem 2.3.2. (Algebraic Limit Theorem for Riemann-Stieltjes Integrals)
Suppose $f, g \in R_\alpha[a,b]$. Then

(i) $\forall k \in \mathbb{R} \quad kf \in R_\alpha[a,b]$ with $\int_a^b kf d\alpha = k\int_a^b f d\alpha$.

(ii) $f+g \in R_\alpha[a,b]$ with $\int_a^b f+g d\alpha = \int_a^b f d\alpha + \int_a^b g d\alpha$.

(iii) $f^2 \in R_\alpha[a,b]$, and $fg \in R_\alpha[a,b]$.

(iv) If $g \neq 0$ on $[a,b]$ and $\frac{1}{g}$ is bounded on $[a,b]$, then $\frac{1}{g} \in R_\alpha[a,b]$ and $\frac{f}{g} \in R_\alpha[a,b]$.

Remark 2.3.1. (1) As a result of (i) and (ii), the set $R_\alpha[a,b]$ is a vector space. Moreover, the function

$$T : R_\alpha[a,b] \to \mathbb{R}$$
$$T(f) = \int_a^b f d\alpha$$

is linear.

(2) The function $g : [0,1] \to \mathbb{R}$ defined by

$$g(x) = \begin{cases} x & 0 < x \leq 1 \\ 1 & x = 0 \end{cases}$$

is an example of a bounded function which is never zero, but $\frac{1}{g}$ is unbounded.

Notation . Let $h : A \subseteq \mathbb{R} \to \mathbb{R}$ be a bounded function.

(i) We denote $\sup\{h(x) : x \in A\}$ by $\sup_A h$ or $\sup_A h(x)$ or $\sup_{x\in A} h(x)$.

(ii) We denote $\inf\{h(x) : x \in A\}$ by $\inf_A h$ or $\inf_A h(x)$ or $\inf_{x\in A} h(x)$.

Lemma 2.3.1. Let $A \subseteq \mathbb{R}$. Let $f : A \to \mathbb{R}$ and $g : A \to \mathbb{R}$ be bounded fucntions. Then

(i) $\sup_A(f+g) \leq \sup_A f + \sup_A g$.

(ii) $\inf_A(f+g) \geq \inf_A f + \inf_A g$.

(iii) $\forall k \geq 0 \quad \sup_A(kf) = k\sup_A f$ and $\inf_A(kf) = k\inf_A f$.

(iv) $\forall k \leq 0 \quad \sup_A(kf) = k\inf_A f$ and $\inf_A(kf) = k\sup_A f$.

(v) $\sup\{|f(x) - f(y)| : x, y \in A\} = \sup_A f - \inf_A f$.

(vi) If $\exists k > 0$ such that $\forall z, w \in A \quad |f(z) - f(w)| \leq k|g(z) - g(w)|$, then

$$\sup_A f - \inf_A f \leq k\left[\sup_A g - \inf_A g\right]$$

Lemma 2.3.2. Let $f : [a,b] \to \mathbb{R}$ and $g : [a,b] \to \mathbb{R}$ be bounded, $\alpha : [a,b] \to \mathbb{R}$ be icreasing, and $P \in \Pi[a,b]$. Then

(i) $U(f+g,\alpha,P) \leq U(f,\alpha,P) + U(g,\alpha,P)$.

(ii) $L(f+g,\alpha,P) \geq L(f,\alpha,P) + L(g,\alpha,P)$.

(iii) $\forall k \geq 0 \;\; U(kf,\alpha,P) = kU(f,\alpha,P)$ and $L(kf,\alpha,P) = kL(f,\alpha,P)$.

(iv) $\forall k < 0 \;\; U(kf,\alpha,P) = kL(f,\alpha,P)$ and $L(kf,\alpha,P) = kU(f,\alpha,P)$.

Proof. Proof of the Algebraic Limit Theorem for Integrals.

(i) $\forall k \in \mathbb{R}, \quad kf \in R_\alpha[a,b]$ with $\int_a^b kf d\alpha = k\int_a^b f d\alpha$:
Note that

$$f \in R_\alpha[a,b] \implies f \text{ is bounded on } [a,b] \implies kf \text{ is bounded on } [a,b].$$

Since $f \in R_\alpha[a,b]$, there exists a sequence $(P_n)n \geq 1$ in $\Pi[a,b]$ such that

$$\lim_{n\to\infty}[U(f,\alpha,P_n) - L(f,\alpha,P_n)] = 0, \int_a^b f d\alpha = \lim_{n\to\infty} U(f,\alpha,P_n) = \lim_{n\to\infty} L(f,\alpha,P_n).$$

In what follows, we'll show that

$$\lim_{n\to\infty}[U(kf,\alpha,P_n) - L(kf,\alpha,P_n)] = 0$$

and so $kf \in R_\alpha[a,b]$. We may consider two cases:

Case 1: $k \geq 0$

$$\begin{aligned}\lim_{n\to\infty}[U(kf,\alpha,P_n) - L(kf,\alpha,P_n)] &= \lim_{n\to\infty}[kU(f,\alpha,P_n) - kL(f,\alpha,P_n)] \\ &= k\lim_{n\to\infty}[U(f,\alpha,P_n) - L(f,\alpha,P_n)] \\ &= 0.\end{aligned}$$

Case 2: $k < 0$

$$\begin{aligned}\lim_{n\to\infty}[U(kf,\alpha,P_n) - L(kf,\alpha,P_n)] &= \lim_{n\to\infty}[kL(f,\alpha,P_n) - kU(f,\alpha,P_n)] \\ &= -k\lim_{n\to\infty}[U(f,\alpha,P_n) - L(f,\alpha,P_n)] \\ &= 0.\end{aligned}$$

Moreover, if $k \geq 0$, then

$$\int_a^b kf d\alpha = \lim_{n\to\infty} U(kf,\alpha,P_n) = \lim_{n\to\infty} k \cdot U(f,\alpha,P_n) = k \cdot \int_a^b f d\alpha.$$

Similarly, for $k < 0$,

$$\int_a^b kf d\alpha = \lim_{n\to\infty} U(kf,\alpha,P_n) = \lim_{n\to\infty} k \cdot L(f,\alpha,P_n) = k \cdot \int_a^b f d\alpha.$$

(ii) $f+g \in R_\alpha[a,b]$ with $\int_a^b f+g d\alpha = \int_a^b f d\alpha + \int_a^b g d\alpha$:
Note that

$$\left.\begin{array}{l} f \in R_\alpha[a,b] \implies f \text{ is bounded on } [a,b] \\ g \in R_\alpha[a,b] \implies g \text{ is bounded on } [a,b] \end{array}\right\} \implies f+g \text{ is bounded on } [a,b].$$

Since $f \in R_\alpha[a,b]$ there exists $(P_n)_{n\geq 1}$ in $\Pi[a,b]$ such that

$$\lim_{n\to\infty}[U(f,\alpha,P_n) - L(f,\alpha,P_n)] = 0, \quad \int_a^b f d\alpha = \lim_{n\to\infty} U(f,\alpha,P_n) = \lim_{n\to\infty} L(f,\alpha,P_n).$$

Since $g \in R_\alpha[a,b]$ there exists $(Q_n)_{n\geq 1}$ in $\Pi[a,b]$ such that

$$\lim_{n\to\infty}[U(g,\alpha,Q_n) - L(g,\alpha,Q_n)] = 0, \quad \int_a^b g d\alpha = \lim_{n\to\infty} U(g,\alpha,Q_n) = \lim_{n\to\infty} L(g,\alpha,Q_n).$$

In what follows, we will show that

$$\lim_{n\to\infty}[U(f+g,\alpha,P_n \cup Q_n) - L(f+g,\alpha,P_n \cup Q_n)] = 0$$

and so $f+g \in R_\alpha[a,b]$. We have

$$\begin{aligned} L(f+g,\alpha,P_n\cup Q_n) &\geq L(f,\alpha,P_n\cup Q_n)+L(g,\alpha,P_n\cup Q_n)\\ &\geq L(f,\alpha,P_n)+L(g,\alpha,Q_n)\end{aligned}$$

and

$$\begin{aligned} U(f+g,\alpha,P_n\cup Q_n) &\leq U(f,\alpha,P_n\cup Q_n)+U(g,\alpha,P_n\cup Q_n)\\ &\leq U(f,\alpha,P_n)+U(g,\alpha,Q_n)\end{aligned}$$

so

$$\begin{aligned} L(f,\alpha,P_n)+L(g,\alpha,Q_n) &\leq L(f+g,\alpha,P_n\cup Q_n)\\ &\leq U(f+g,\alpha,P_n\cup Q_n)\\ &\leq U(f,\alpha,P_n)+U(g,\alpha,Q_n).\end{aligned}$$

Therefore,

$$\begin{aligned} 0 &\leq U(f+g,\alpha,P_n\cup Q_n)-L(f+g,\alpha,P_n\cup Q_n)\\ &\leq [U(f,\alpha,P_n)-L(f,\alpha,P_n)]+[U(f,\alpha,Q_n)-L(f,\alpha,Q_n)]\to 0+0=0.\end{aligned}$$

By the squeeze theorem,

$$\lim_{n\to\infty}[U(f+g,\alpha,P_n\cup Q_n)-L(f+g,\alpha,P_n\cup Q_n)]=0.$$

Moreover,

$$\begin{aligned} \int_a^b f+g\,d\alpha &= \lim_{n\to\infty}[U(f+g,\alpha,P_n\cup Q_n)]\\ &\leq \lim_{n\to\infty}[U(f,\alpha,P_n)+U(g,\alpha,P_n)]=\int_a^b f\,d\alpha+\int_a^b g\,d\alpha\\ \int_a^b f+g\,d\alpha &= \lim_{n\to\infty}[L(f+g,\alpha,P_n\cup Q_n)]\\ &\geq \lim_{n\to\infty}[L(f,\alpha,P_n)+L(g,\alpha,P_n)]=\int_a^b f\,d\alpha+\int_a^b g\,d\alpha.\end{aligned}$$

This implies $\int_a^b f+g\,d\alpha=\int_a^b f\,d\alpha+\int_a^b g\,d\alpha$.

(iii) $f^2\in R_\alpha[a,b]$:
Note that $\phi:\mathbb{R}\to\mathbb{R}$, $\phi(x)=x^2$ is continuous, so since $f\in R_\alpha[a,b]$, $\phi\circ f\in R_\alpha[a,b] \implies f^2\in R_\alpha[a,b]$.
$fg\in R_\alpha[a,b]$:
Note that

$$\left.\begin{aligned}(f+g)^2&=f^2+2fg+g^2\\(f-g)^2&=f^2-2fg+g^2\end{aligned}\right\}\implies 4fg=(f+g)^2-(f-g)^2\implies fg=\frac{1}{4}\left[(f+g)^2-(f-g)^2\right].$$

Therefore,

$$\begin{aligned} f,g\in R_\alpha[a,b] &\overset{(i)}{\implies} f,g,-g\in R_\alpha[a,b]\\ &\overset{(ii)}{\implies} f+g,f-g\in R_\alpha[a,b]\\ &\overset{(iii)}{\implies} (f+g)^2,(f-g)^2\in R_\alpha[a,b]\\ &\overset{(i)}{\implies} (f+g)^2,-(f-g)^2\in R_\alpha[a,b]\\ &\overset{(ii)}{\implies} (f+g)^2-(f-g)^2\in R_\alpha[a,b]\\ &\overset{(i)}{\implies} \frac{1}{4}\left[(f+g)^2-(f+g)^2\right]\in R_\alpha[a,b]\\ &\implies fg\in R_\alpha[a,b].\end{aligned}$$

□

Theorem 2.3.3. (Order Properties of Riemann-Stieltjes Integrals)
Supose $f, g \in R_\alpha[a,b]$. Then

(i) If $m \le f(x) \le M \quad \forall x \in [a,b]$, then $m(\alpha(b)-\alpha(a)) \le \int_a^b f d\alpha \le M(\alpha(b)-\alpha(a))$.

(ii) If $f \le g$ on $[a,b]$, then $\int_a^b f d\alpha \le \int_a^b g d\alpha$.

Proof. (i) Note that for any $P \in \Pi[a,b]$,

$$\int_a^b f d\alpha = L(f,\alpha) \ge L(f,\alpha,P)$$
$$\int_a^b f d\alpha = U(f,\alpha) \le U(f,\alpha,P).$$

In particular, for the partition $P = \{a,b\}$, we have

$$\int_a^b f d\alpha \ge L(f,\alpha,P) = \inf\{f(x) : x \in [a,b]\}(\alpha(b)-\alpha(a)) \ge m(\alpha(b)-\alpha(a))$$
$$\int_a^b f d\alpha \le U(f,\alpha,P) = \sup\{f(x) : x \in [a,b]\}(\alpha(b)-\alpha(a)) \le M(\alpha(b)-\alpha(a))$$

(ii) Let $h = g - f$. We have $h \ge 0$, so by part (i)

$$\begin{aligned} 0(\alpha(b)-\alpha(a)) &\le \int_a^b h d\alpha \\ &= \int_a^b g d\alpha - \int_a^b f d\alpha \\ \implies \int_a^b f d\alpha &\le \int_a^b g d\alpha. \end{aligned}$$

□

Theorem 2.3.4. ($|\int_a^b f d\alpha| \le \int_a^b |f| d\alpha$)
Suppose $f \in R_\alpha[a,b]$. Then

(i) $|f| \in R_\alpha[a,b]$.

(ii) $|\int_a^b f d\alpha| \le \int_a^b |f| d\alpha$.

Proof. (i) Since $\phi : \mathbb{R} \to \mathbb{R}$, $\phi(x) = |x|$ is continuous and f is Riemann-Stieltjes integrable, it follows that $|f| = \phi \circ f$ is Riemann-Stieltjes integrable. Hence $|f| \in R_\alpha[a,b]$.

(ii) Recall that $|t| \le s \iff -s \le t \le s$. So our goal is to show that

$$-\int_a^b |f| d\alpha \le \int_a^b f d\alpha \le \int_a^b |f| d\alpha$$

Clearly, (for any c, $-|c| \le c \le |c|$)

$$\begin{aligned} -|f(x)| \le f(x) \le |f(x)| \quad \forall x \in [a,b] \\ \implies -\int_a^b |f| d\alpha \le \int_a^b f d\alpha \le \int_a^b |f| d\alpha \quad \forall x \in [a,b] \end{aligned}$$

□

Theorem 2.3.5. (The Mean Value Theorem for Riemann-Stieltjes Integrals)
Suppose $f : [a,b] \to \mathbb{R}$ is continuous, $\alpha : [a,b] \to \mathbb{R}$ is increasing, and $\alpha(a) \ne \alpha(b)$. Then

$$\exists c \in [a,b] \text{ such that } f(x) = \frac{\int_a^b f d\alpha}{\alpha(b)-\alpha(a)}$$

Proof. Because f is continuous on $[a,b]$, and $[a,b]$ is compact, by the Extreme Value Theorem f attains its maximum and minimum on $[a,b]$. Let $m = \min\{f(x) : x \in [a,b]\}$ and $M = \max\{f(x) : x \in [a,b]\}$. We have

$$\begin{aligned} &\forall x \in [a,b] \quad m \le f(x) \le M \\ &\implies m(\alpha(b) - \alpha(a)) \le \int_a^b f d\alpha \le M(\alpha(b) - \alpha(a)) \\ &\implies m \le \frac{\int_a^b f d\alpha}{\alpha(b) - \alpha(a)} \le M. \end{aligned}$$

Since f is continuous, it follows from the Intermediate Value Theorem

$$\exists c \in [a,b] \text{ such that } f(c) = \frac{\int_a^b f d\alpha}{\alpha(b) - \alpha(a)}$$

□

Theorem 2.3.6. (Generalized Mean Value Theorem for Riemann-Stieltjes Integrals) Suppose $f : [a,b] \to \mathbb{R}$ is continuous, $\alpha : [a,b] \to \mathbb{R}$ is increasing, $g \in R_\alpha[a,b]$, and either $g \ge 0$ on $[a,b]$ or $g \le 0$ on $[a,b]$. Then

$$\exists c \in [a,b] \text{ such that } \int_a^b f g d\alpha = f(c) \int_a^b g d\alpha.$$

Proof. We will prove the claim for the case $g \ge 0$ on $[a,b]$. The proof for the case $g \le 0$ is analogous. Since f is continuous on $[a,b]$ and $[a,b]$ is compact, f contains its maximum and minimum on $[a,b]$. Let $M = \max\{f(x) : x \in [a,b]\}$ and $m = \min\{f(x) : x \in [a,b]\}$. Then, since $g \le 0$ on $[a,b]$,

$$\begin{aligned} &\forall x \in [a,b] \quad m \le f(x) \le M \\ \implies &\forall x \in [a,b] \quad mg(x) \le f(x)g(x) \le Mg(x) \\ \implies &\forall x \in [a,b] \quad m \int_a^b g d\alpha \le \int_a^b f g d\alpha \le M \int_a^b g d\alpha. \end{aligned}$$

Now, we may consider two cases:

Case 1: $\int_a^b g d\alpha = 0$ If $\int_a^b g d\alpha = 0$, it follows that $0 \le \int_a^b f g d\alpha \le 0$ and so $\int_a^b f g d\alpha = 0$. So in this case, the claim holds for every $c \in [a,b]$:

$$\int_a^b f g d\alpha = f(c) \int_a^b g d\alpha.$$

Case 2: $\int_a^b g d\alpha \ne 0$ In this case, since $g > 0$, it follows that $\int_a^b g d\alpha > 0$ and

$$m \le \frac{\int_a^b f g d\alpha}{\int_a^b g d\alpha} \le M.$$

Since f is continuous, it follows from the Intermediate Value Theorem that

$$\exists c \in [a,b] \text{ such that } f(x) = \frac{\int_a^b f g d\alpha}{\int_a^b g d\alpha}.$$

That is, $\int_a^b f g d\alpha = f(c) \int_a^b g d\alpha$.

□

Theorem 2.3.7. (Additivity for Riemann-Stieltjes Integrals) Suppose $f : [a,b] \to \mathbb{R}$ is continuous, $\alpha : [a,b] \to \mathbb{R}$ is increasing, and $c \in (a,b)$. Then

$$f \in R_\alpha[a,b] \iff f \in R_\alpha[a,c] \text{ and } f \in R_\alpha[c,b].$$

In this case, we have

$$\int_a^b f d\alpha = \int_a^c f d\alpha + \int_c^b f d\alpha.$$

□

2.4 The Fundamental Theorem of Calculus and Techniques of Integration

Lemma 2.4.1.
Suppose $f : [a,b] \to \mathbb{R}$ is bounded, $\alpha : [a,b] \to \mathbb{R}$ is increasing, $P = \{x_0, ..., x_n\} \in \Pi[a,b]$, and $R \in \mathbb{R}$.

(i) If for all tags $(s_k)_{a \le k \le n}$ of P we have $\sum_{k=1}^n f(s_k)\Delta\alpha_k \le R$, then $U(f,\alpha,P) \le R$.

(ii) If for all tags $(s_k)_{a \le k \le n}$ of P we have $R \le \sum_{k=1}^n f(s_k)\Delta\alpha_k$, then $L(f,\alpha,P) \ge R$.

Proof. (i) If α is constant, then $\sum_{k=1}^n f(s_k)\Delta\alpha_k = 0$ and $U(f,\alpha,P) = \sum_{k=1}^n M_k\Delta\alpha_k = 0$, so the claim obviously holds. Suppose $\alpha(b) \neq \alpha(a)$. It's enough to show

$$\forall \epsilon > 0 \;\; U(f,\alpha,P) \le R + \epsilon.$$

Let $\epsilon > 0$ be given. For each $k \in \{1, ..., n\}$ we have

$$M_k = \sup\{f(x) : x \in [x_{k-1}, x_k]\} \implies \exists s_k \in [x_{k-1}, x_k] \text{ such that } M_k - \frac{\epsilon}{\alpha(b)-\alpha(a)} < f(s_k).$$

We have

$$\begin{aligned} U(f,\alpha,P) &= \sum_{k=1}^n M_k\Delta\alpha_k < \sum_{k=1}^n \left[f(s_k) + \frac{\epsilon}{\alpha(b)-\alpha(a)}\right]\Delta\alpha_k \\ &= \sum_{k=1}^n f(s_k)\Delta\alpha_k + \frac{\epsilon}{\alpha(b)-\alpha(a)}\sum_{k=1}^n \Delta\alpha_k \\ &\le R + \epsilon \end{aligned}$$

as desired.

(ii) Completely analogous to (i). □

Theorem 2.4.1.
Suppose $f : [a,b] \to \mathbb{R}$ is bounded, $\alpha[a,b] \to \mathbb{R}$ is increasing, and $\alpha' \in R[a,b]$. Then

$$f \in R_\alpha[a,b] \iff f\alpha' \in R[a,b]$$

and in this case,

$$\int_a^b f d\alpha = \int_a^b f(x)\alpha'(x)dx.$$

Proof. It is enough to show that

$$\begin{aligned} U(f,\alpha) &= U(f\alpha') \\ L(f,\alpha) &= L(f\alpha') \end{aligned} \tag{$*$}$$

Indeed, if we prove $(*)$, then we can write

$$\begin{aligned} f \in R_\alpha[a,b] &\iff U(f,\alpha) = L(f,\alpha) \\ &\iff U(f\alpha') = L(f\alpha') \\ &\iff f\alpha' \in R[a,b]. \end{aligned}$$

Moreover,

$$\int_a^b f d\alpha = U(f,\alpha) \overset{(*)}{=} U(f\alpha') = \int_a^b f(x)\alpha'(x)dx.$$

In what follows, we will prove $U(f,\alpha) = U(f\alpha')$. The proof for $L(f,\alpha) = L(f\alpha')$ is completely analogous.

let $P = \{x_0, ..., x_n\}$ be any partition of $[a, b]$. Let (s_k) be any tag of P. We have

$$\begin{aligned}
\left|\sum_{k=1}^{n} f(s_k)\Delta\alpha_k - \sum_{k=1}^{n} f(s_k)\alpha'(s_k)\Delta x_k\right| &= \left|\sum_{k=1}^{n} f(s_k)\alpha'(t_k)\Delta x_k - \sum_{k=1}^{n} f(s_k)\alpha'(s_k)\Delta x_k\right| \qquad \text{(MVT on } \Delta\alpha_k\text{)} \\
&= \left|\sum_{k=1}^{n} f(s_k)\left(\alpha'(t_k) - \alpha'(s_k)\right)\Delta x_k\right| \\
&\leq \sum_{k=1}^{n} |f(s_k)| \cdot |\alpha'(t_k) - \alpha'(s_k)|\,\Delta x_k \\
&\leq \hat{M}\sum_{k=1}^{n} |\alpha'(t_k) - \alpha'(s_k)|\,\Delta x_k \qquad (\hat{M} = \sup_{a\leq x\leq b}|f(x)|) \\
&\leq \hat{M}\sum_{k=1}^{n}\left[\sup_{I_k}\alpha' - \inf_{I_k}\alpha'\right]\Delta x_k \qquad \text{(Lemma 2.2.1)} \\
&= \hat{M}\left[U(\alpha', P) - L(\alpha', P)\right]
\end{aligned}$$

So,

$$\left|\sum_{k=1}^{n} f(s_k)\Delta\alpha_k - \sum_{k=1}^{n} f(s_k)\alpha'(s_k)\Delta x_k\right| \leq \hat{M}\left[U(\alpha', P) - L(\alpha', P)\right].$$

Therefore,

$$\begin{aligned}
(1)\quad \sum_{k=1}^{n} f(s_k)\Delta\alpha_k &\leq \sum_{k=1}^{n} f(s_k)\alpha'(s_k)\Delta x_k + \hat{M}\left[U(\alpha', P) - L(\alpha', P)\right] \\
&\leq U(f\alpha', P) + \hat{M}\left[U(\alpha', P) - L(\alpha', P)\right].
\end{aligned}$$

So, by Lemma 2.4.1,

$$U(f, \alpha, P) \leq U(f\alpha', P) + \hat{M}\left[U(\alpha', P) - L(\alpha', P)\right].$$

$$(2)\quad \sum_{k=1}^{n} f(s_k)\alpha'(s_k)\Delta x_k \leq \sum_{k=1}^{n} f(s_k)\Delta\alpha_k + \hat{M}\left[U(\alpha', P) - L(\alpha', P)\right]$$

So, by Lemma 2.4.1,

$$U(f\alpha', P) \leq U(f, \alpha, P) + \hat{M}\left[U(\alpha', P) - L(\alpha', P)\right].$$

It follows from (1) and (2) that

$$(3)\quad |U(f, \alpha, P) - U(f\alpha', P)| \leq \hat{M}\left[U(\alpha', P) - L(\alpha', P)\right].$$

Note that

$$\begin{aligned}
U(f,\alpha) = \inf\{U(f,\alpha,P) : P \in \Pi\} &\implies \exists(P_n^{(1)}) \subseteq \Pi \text{ such that } \lim_{n\to\infty} U(f,\alpha,P_n^{(1)}) = U(f,\alpha) \\
U(f\alpha') = \inf\{U(f\alpha',P) : P \in \Pi\} &\implies \exists(P_n^{(2)}) \subseteq \Pi \text{ such that } \lim_{n\to\infty} U(f\alpha',P_n^{(2)}) = U(f\alpha') \\
\alpha' \in R[a,b] &\implies \exists(P_n^{(3)}) \subseteq \Pi \text{ such that } \lim_{n\to\infty}\left[U(\alpha',P_n^{(3)}) - L(\alpha',P_n^{(3)})\right] = 0.
\end{aligned}$$

For each $n \in \mathbb{N}$, let $P_n = P_n^{(1)} \cup P_n^{(2)} \cup P_n^{(3)}$. By the Squeeze Theorem,

$$\begin{aligned}
\forall n \geq 1 \;\; U(f,\alpha) \leq U(f,\alpha,P_n) \leq U(f,\alpha,P_n^{(1)}) &\implies \lim_{n\to\infty} U(f,\alpha,P_n) = U(f,\alpha) \\
\forall n \geq 1 \;\; U(f\alpha') \leq U(f\alpha',P_n) \leq U(f\alpha',P_n^{(2)}) &\implies \lim_{n\to\infty} U(f\alpha',P_n) = U(f\alpha').
\end{aligned}$$

Since P_n is a refinement of $P_n^{(3)}$,

$$0 \leq [U(\alpha', P_n) - L(\alpha', P_n)] \leq U(\alpha', P_n^{(3)}) - L(\alpha', P_n^{(3)}) \implies \lim_{n\to\infty}[U(\alpha', P_n) - L(\alpha', P_n)] = 0.$$

It follows from (3) that

$$\forall n \geq 1 \;\; 0 \leq |U(f,\alpha,P_n) - U(f\alpha',P_n)| \leq \hat{M}\left[U(\alpha',P_n) - L(\alpha',P_n)\right].$$

So, by the Squeeze Theorem,

$$\begin{aligned}
&\lim_{n\to\infty} |U(f,\alpha,P_n) - U(f\alpha',P_n)| = 0 \\
\implies &\left|\lim_{n\to\infty} (U(f,\alpha,P_n) - U(f\alpha',P_n))\right| = 0 \\
\implies &|U(f,\alpha) - U(f\alpha')| = 0 \\
\implies &U(f,\alpha) - U(f\alpha') = 0 \\
\implies &U(f,\alpha) = U(f\alpha')
\end{aligned}$$

□

Theorem 2.4.2. (Change of Variables for Integrals)
Let $f \in R_\alpha[a,b]$, and let $\phi : [A,B] \to [a,b]$ be onto and strictly increasing. If we let $g = f \circ \phi$ and $\beta = \alpha \circ \phi$, then

$$g \in R_\beta[A,B] \text{ and } \int_a^b f d\alpha = \int_A^B g d\beta.$$

Remark 2.4.1. Consider the special case where $\alpha(x) = x$ and $\phi' \in \mathbb{R}$. In this case, $\beta = \alpha \circ \phi = \phi$. So, the conclusion of the theorem can be written as

$$\int_a^b f(x)dx = \int_A^B (f \circ \phi)d\phi = \int_A^B f(\phi(y))\phi'(y)dy.$$

Proof. First, note that

$$\left.\begin{array}{l} \alpha : [a,b] \to \mathbb{R} \text{ is increasing} \\ \phi : [A,B] \to \mathbb{R} \text{ is increasing} \end{array}\right\} \implies \beta = \alpha \circ \phi \text{ is increasing on } [A,B]$$

and

$$\left.\begin{array}{l} f \in R_\alpha[a,b] \implies f \text{ is bounded on } [a,b] \\ \forall x \in [A,B]\ a \le \phi(x) \le b \implies \phi \text{ is bounded on } [A,B] \end{array}\right\} \implies g = f \circ \phi \text{ is bounded on } [A,B].$$

Also, note that there is a 1-1 correspondance between $\Pi[a,b]$ and $\Pi[A,B]$:

$$\begin{aligned} H : \Pi[a,b] &\to \Pi[A,B] \\ P = \{x_0,...,x_n\} &\mapsto Q = \{y_0,,...,y_n\} \quad (x_k = \phi(y_k)) \end{aligned}$$

Under this 1-1 correspondance, we have

$$\begin{aligned}
&\forall 1 \le k \le n \quad \phi([y_{k-1},y_k]) = [x_{k-1},x_k] \\
&\forall 1 \le k \le n \quad \Delta\beta_k = \beta(y_k) - \beta(y_{k-1}) = \alpha(\phi(y_k)) - \alpha(\phi(y_{k-1})) = \alpha(x_k) - \alpha(x_{k-1}) = \Delta\alpha_k \\
&\forall 1 \le k \le n \quad M_k^{(g)} = \sup\{g(y) : y \in [y_{k-1},y_k]\} = \sup\{f(x) : x \in [x_{k-1},x_k]\} = M_k^{(f)} \\
&\forall 1 \le k \le n \quad m_k^{(g)} = \inf\{g(y) : y \in [y_{k-1},y_k]\} = \inf\{f(x) : x \in [x_{k-1},x_k]\} = m_k^{(f)}.
\end{aligned}$$

Thus, under the above correspondace,

$$\begin{aligned} U(f,\alpha,P) &= U(g,\beta,Q) \\ L(f,\alpha,P) &= L(f,\alpha,Q) \end{aligned}$$

In order to show $g \in R_\beta[A,B]$, by the Cauchy Criterion, it is enough to prove that

$$\forall \epsilon > 0 \quad \exists Q \in \Pi[A,B] \text{ such that } U(g,\beta,Q) - L(g,\beta,Q) < \epsilon.$$

Let $\epsilon > 0$ be given. Since $f \in R_\alpha[a,b]$, $\exists P \in \Pi[a,b]$ such that

$$U(f,\alpha,P) - L(f,\alpha,P) < \epsilon.$$

We claim that $Q = H(P)$ is the partition that we're looking for. Indeed,

$$U(g,\beta,H(P)) - L(g,\beta,H(P)) = U(f,\alpha,P) - L(f,\alpha,P) < \epsilon.$$

Also,

$$\begin{aligned}\int_a^b f d\alpha = L(f,\alpha) &= \sup\{L(f,\alpha,P) : P \in \Pi[a,b]\} \\ &= \sup\{L(g,\beta,H(P)) : H(P) \in \Pi[A,B]\} \\ &= L(g,\beta) \\ &= \int_A^B g d\beta.\end{aligned}$$

□

Theorem 2.4.3. (The Fundamental Theorem of Calculus)

(i) Supose $f \in R[a,b]$ and $F : [a,b] \to \mathbb{R}$ satisfies $F'(x) = f(x) \ \ \forall x \in [a,b]$. Then

$$\int_a^b f = F(b) - F(a).$$

(ii) Suppose $g \in R[a,b]$ and $G : [a,b] \to \mathbb{R}$ is defined by $G(x) = \int_a^x g(t)dt$. Then

(1) G is uniformly continuous on $[a,b]$.

(2) If g is differentiable at $c \in [a,b]$, then G is differentiable at c and

$$G'(c) = g(c).$$

Proof. (i) In what follows, we'll show that

$$\forall P \in \Pi[a,b] \ \ L(f,P) \le F(b) - F(a) \le U(f,P). \qquad (*)$$

As a consequence of this:

(I) $F(b) - F(a)$ is an upper bound for $\{L(f,P) : P \in \Pi\}$, so

$$\sup\{L(f,P) : P \in \Pi\} \le F(b) - F(a) \implies L(f) \le F(b) - F(a)$$

(II) $F(b) - F(a)$ is a lower bound for $\{U(f,P) : P \in \Pi\}$, so

$$\inf\{U(f,P) : P \in \Pi\} \ge F(b) - F(a) \implies U(f) \ge F(b) - F(a).$$

Thus, $L(f) \le F(b) - F(a) \le U(f)$. Since $f \in R[a,b]$, $U(f) = L(f) = \int_a^b f d\alpha$, giving us

$$\int_a^b f d\alpha \le F(b) - F(a) \le \int_a^b f d\alpha.$$

Therefore, $\int_a^b f d\alpha$.

So, we just need to prove $(*)$. Let $P = \{x_0, ..., x_n\} \in \Pi[a,b]$. We have

$$\begin{aligned}F(b) - F(a) &= \sum_{k=1}^n F(x_k) - F(x_{k-1}) \\ &= \sum_{k=1}^n F'(t_k)(x_k - x_{k-1}) \text{for some } t_k \in [x_{k-1}, x_k] \qquad \text{(MVT for derivatives)} \\ &= \sum_{k=1}^n f(t_k)\Delta x_k.\end{aligned}$$

Therefore, it follows from

$$L(f,p) = \sum_{k=1}^n m_k \Delta x_k \le \sum_{k=1}^n f(t_k)\Delta x_k \le \sum_{k=1}^n M_k \Delta x_k = U(f,P)$$

that

$$L(f,P) \le F(b) - F(a) \le U(f,P).$$

(ii) (1) Our goal is to show G is uniformly continuous on $[a,b]$. That is, we need to show

$$\forall \epsilon > 0 \ \exists \delta > 0 \text{ such that } \forall x, y \in [a,b] \text{ if } |x-y| < \delta, \text{ then } |G(x) - G(y)| < \epsilon.$$

Let $\epsilon > 0$ be given. Our goal is to find $\delta > 0$ such that

$$\forall x, y \in [a,b] \text{ if } |x-y| < \delta, \text{ then } |G(x) - G(y)| < \epsilon. \qquad (*)$$

Let $x, y \in [a,b]$. If $x \geq y$,

$$\begin{aligned} |G(x) - G(y)| &= \left| \int_a^x g(t)dt - \int_a^y g(t)dt \right| \\ &= \left| \int_y^x g(t)dt \right| \\ &\leq \int_y^x |g(t)|dt \\ &\leq \int_y^x R dt \qquad \text{(Let } R = \sup_a \leq t \leq b |g|) \\ &= R(x-y) \\ &= R|x-y|. \end{aligned}$$

If $x < y$, then

$$\begin{aligned} |G(x) - G(y)| &= |G(y) - G(x)| \\ &= \left| \int_a^y g(t)dt - \int_a^x g(t)dt \right| \\ &= \left| \int_x^y g(t)dt \right| \\ &\leq \int_x^y |g(t)|dt \\ &\leq \int_x^y R dt \\ &= R(y-x) \\ &= R|x-y|. \end{aligned}$$

Thus, $\forall x, y \in [a,b]$

$$|G(x) - G(y)| \leq R|x-y|.$$

Hence, to ensure $|G(x) - G(y)| < \epsilon$ it is enough to make $R|x-y| < \epsilon$. That is, it is enough to make $|x-y| < \epsilon/R$. Choosing $\delta = \epsilon/R$ does the job, and so $(*)$ is satisfied.

(2) Suppose g is continuous at $c \in [a,b]$. Our goal is to show $G'(c) = g(c)$. That is, we want to show

$$\lim_{x \to c} \frac{G(x) - G(c)}{x - c} = g(c).$$

That is, our goal is to show

$$\forall \epsilon > 0 \ \exists \delta > 0 \text{ such that } \text{ if } 0 < |x-c| < \delta, \text{ then } \left| \frac{G(x) - G(c)}{x-c} - g(c) \right| < \epsilon.$$

Let $\epsilon > 0$ be given. Since g is continuous at c, for the given ϵ, there exists $\hat{\delta} > 0$ such that

$$\text{if } \underset{t \in [a,b]}{|t-c| < \hat{\delta}}, \text{ then } |g(t) - g(c)| < \epsilon.$$

We claim that this $\hat{\delta}$ can be used as the δ we're looking for. Indeed, if we let $\delta = \hat{\delta}$, then

Case 1: $0 < |x-c| < \hat{\delta}, x \in [a,b], x > c$

$$\begin{aligned}
\left|\frac{G(x)-G(c)}{x-c} - g(c)\right| &= \left|\frac{G(x)-G(c)-g(c)(x-c)}{x-c}\right| \\
&= \left|\frac{\int_a^x g(t)dt - \int_a^c g(t)dt - \int_a^x g(c)dt}{x-c}\right| \\
&= \left|\frac{\int_c^x g(t)dt - \int_c^x g(c)dt}{x-c}\right| \\
&= \left|\frac{\int_c^x g(t)-g(c)dt}{x-c}\right| \\
&= \frac{1}{|x-c|}\left|\int_c^x g(t)-g(c)dt\right| \\
&\leq \frac{1}{|x-c|}\int_c^x |g(t)-g(c)|dt \\
&\leq \frac{1}{x-c}\int_c^x \frac{\epsilon}{2}dt \\
&= \frac{\epsilon/2}{x-c}(x-c) < \epsilon.
\end{aligned}$$

Case 2: $0 < |x-c| < \hat{\delta}, x \in [a,b], x < c$

$$\begin{aligned}
\left|\frac{G(x)-G(y)}{x-c} - g(c)\right| &= \left|\frac{G(x)-G(y)-g(c)(x-c)}{x-c}\right| \\
&= \left|\frac{g(c)(c-x)-(G(c)-G(x))}{c-x}\right| \\
&= \left|\frac{\int_x^c g(c)dt - \int_a^c g(t)dt + \int_a^x g(t)dt}{c-x}\right| \\
&\vdots \\
&= \left|\frac{\int_x^c g(c)-g(t)dt}{c-x}\right| \\
&\leq \frac{1}{c-x}\int_x^c |g(c)-g(t)|dt \\
&\leq \frac{\int_x^c \frac{\epsilon}{2}dt}{c-x} \\
&< \epsilon.
\end{aligned}$$

□

Notation . Let I be a nonempty interval in $\mathbb{R}$.

$$C^0(I) = C(I) = \{f : I \to \mathbb{R} \mid f \text{ is continuous}\}$$

$$\forall n \geq 1 \quad C^n(I) = \left\{f : I \to \mathbb{R} \mid f, f', ..., f^{(n)} \text{ exist and are continuouts}\right\}$$

Theorem 2.4.4. (Integration by Parts)
Suppose $u : [a,b] \to \mathbb{R}$ and $v : [a,b] \to \mathbb{R}$ are differentiable and $u', v' \in R[a,b]$. Then

(i) $uv' \in R[a,b]$.

(ii) $u'v \in R[a,b]$.

(iii) $\int_a^b uv'dx = u(b)v(b) - u(a)v(a) - \int_a^b u'vdx$.

Proof. (i) $u : [a,b] \to \mathbb{R}$ is differentiable, so

$$\left.\begin{array}{r} u \in C[a,b] \implies u \in R[a,b] \\ u \in R[a,b] \\ v' \in R[a,b] \end{array}\right\} \implies uv' \in R[a,b]$$

(ii) Analogous to (i).

(iii) By the product rule,

$$(uv)' = u'v + uv'.$$

In particular, since $(uv)'$ is the sum of integrable functions it follows that $(uv)' \in R[a,b]$. Integrate both sides:

$$\int_a^b (uv)'dx = \int_a^b u'vdx = \int_a^b uv'dx \qquad (I)$$

By the Fundamental Theorem of Calculus part (i),

$$\int_a^b (uv)'dx = [uv]_{x=a}^{x=b} = u(b)v(b) - u(a)v(a) \qquad (II)$$

By (I) and (II), we have

$$\begin{aligned} u(b)v(b) - u(a)v(a) &= \int_a^b u'vdx + \int_a^b uv'dx \\ \implies \int_a^b u'vdx &= u(b)v(b) - u(a)v(a) - \int_a^b uv'dx \end{aligned}$$

□

Definition 2.4.1. (Unit Step Function)
The unit step function $I : \mathbb{R} \to \mathbb{R}$ is defined by

$$I(x) = \begin{cases} 0 & \text{if } x \leq 0 \\ 1 & \text{if } x > 0 \end{cases}.$$

Note that for all $s \in \mathbb{R}$,

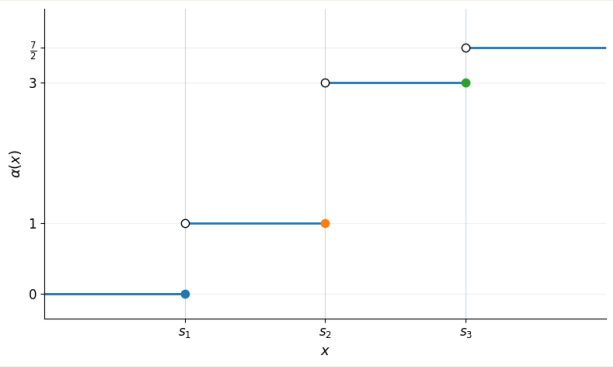
$$I(x-s) = \begin{cases} 0 & \text{if } x \leq s \\ 1 & \text{if } x > 2 \end{cases}.$$

Also, for all $c \neq 0$,

$$cI(x-s) = \begin{cases} 0 & \text{if } x \leq s \\ 1 & \text{if } x > s \end{cases}.$$

Example 2.4.1. Suppose $s_1 < s_2 < s_3$. Graph the function

$$\begin{aligned} \alpha(x) &= I(x-s_1) + 2I(x-s_2) + \frac{1}{2}I(x-s_3) \\ \implies \alpha(x) &= \begin{cases} 0+0+0 & \text{if } x \leq s_1 \\ 1+0+0 & \text{if } s_1 < x \leq s_2 \\ 1+2+0 & \text{if } s_2 < x \leq s_3 \\ 1+2+\frac{1}{2} & \text{if } s_3 < x \end{cases} \end{aligned}$$



Theorem 2.4.5.
Suppose $f : [a, b] \to \mathbb{R}$ is bounded, $s \in (a, b)$, f is right continuous at s, and $\alpha(x) = I(x - s)$. Then $f \in R_\alpha[a, b]$ and $\int_a^b f d\alpha = f(s)$.

Proof. In what follows, we will introduce a sequence of partitions (P_n) in $\Pi[a, b]$ such that

$$\lim_{n\to\infty} [U(f, \alpha, P_n) - L(f, \alpha, P_n)] = 0 \text{ and } \lim_{n\to\infty} U(f, \alpha, P_n) = \lim_{n\to\infty} L(f, \alpha, P_n) = f(s).$$

This, according to the sequential criterion of integrability, tells us that

$$f \in R_\alpha[a, b] \text{ and } \int_a^b f d\alpha = f(s).$$

Let N be large enough so that $s + \frac{1}{N} < b$. For each $n \geq N$, let

$$P_n = \left\{a, s, s + \frac{1}{n}, b\right\}.$$

We have

$$U(f, \alpha, P_n) = \sum_{k=1}^{3} M_k \Delta\alpha_k = M_2 \Delta\alpha_2 = \left[\sup_{s\leq x\leq s+\frac{1}{n}} f(x)\right](1) = \sup_{s\leq x\leq s+\frac{1}{n}} f(x)$$

$$L(f, \alpha, P_n) = \sum_{k=1}^{3} m_k \Delta\alpha_k = m_2 \Delta\alpha_2 = \left[\inf_{s\leq x\leq s+\frac{1}{n}} f(x)\right](1) = \inf_{s\leq x\leq s+\frac{1}{n}} f(x).$$

Therefore,

$$\lim_{n\to\infty} U(f, \alpha, P_n) = \lim_{n\to\infty} \sup_{s\leq x\leq s+\frac{1}{n}} f(x) = f(x)$$

$$\lim_{n\to\infty} L(f, \alpha, P_n) = \lim_{n\to\infty} \inf_{s\leq x\leq s+\frac{1}{n}} f(x) = f(x).$$

Here, we are using the fact that f is continuous at s, but right continuity would have been enough. We conclude that $\lim_{n\to\infty} [U(f, \alpha, P_n) - L(f, \alpha, P_n)] = f(s) - f(s) = 0$, so $f \in R_\alpha[a, b]$ and $\int_a^b f d\alpha = \lim U(f, \alpha, P_n) = f(s)$. □

Theorem 2.4.6.
Suppose $\forall n \geq 1, c_n \geq 0$, $\sum_{n=1}^{\infty} c_n$ converges. Let $s_1 < s_2 < \ldots$ be points in $(a, b), \alpha : [a, b] \to \mathbb{R}, \alpha(x) = \sum_{n=1}^{\infty} c_n I(x - s_n)$, and $f : [a, b] \to \mathbb{R}$ be continuous. Then

$$f \in R_\alpha[a, b] \text{ and } \int_a^b f d\alpha = \sum_{n=1}^{\infty} c_n f(s_n).$$

Chapter 3

Sequences of Functions

3.1 Sequences of Functions and Pointwise Convergence

Definition 3.1.1. (Sequences of Functions)
Let $A \neq$. If $\forall n \in \mathbb{N}$ we have $f_n : A \to \mathbb{R}$, then

$$f_1, f_2, f_3, \ldots$$

is a sequence of functions and is denoted

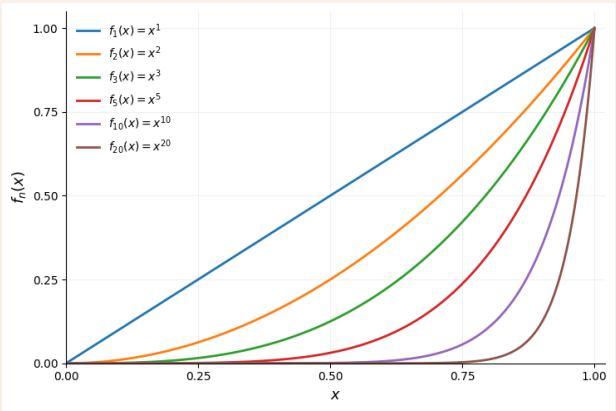
$$(f_n)_{n\geq 1}, (f_n), \text{ or } (f_n : A \to \mathbb{R})_{n\geq 1}.$$

Example 3.1.1. (i) $(f_n)_{n\geq 1} = \left(\frac{\sin(nx)}{n}\right)_{n\geq 1}$ on the set $\mathbb{R}$:

$$\begin{aligned} f_1 : \mathbb{R} \to \mathbb{R}, f_1(x) &= \sin x \\ f_2 : \mathbb{R} \to \mathbb{R}, f_2(x) &= \frac{1}{2}\sin 2x \\ f_3 : \mathbb{R} \to \mathbb{R}, f_3(x) &= \frac{1}{3}\sin 3x \\ &\vdots \end{aligned}$$

(ii) $(f_n)_{n\geq 1} = (x^n)$ on the set $[0, 1]$:

$$\begin{aligned} f_1 : [0,1] \to \mathbb{R}, f_1(x) &= x \\ f_2 : [0,1] \to \mathbb{R}, f_2(x) &= x^2 \\ f_3 : [0,1] \to \mathbb{R}, f_3(x) &= x^3 \\ &\vdots \end{aligned}$$



(iii) Let $f : [0, 1] \to \mathbb{R}$ be a given function. For all $n \in \mathbb{N}$, let $f_n(x) = f(x^n)$. Then (f_n) is a sequence of functions.

Remark 3.1.1. Suppose $(f_n : A \to \mathbb{R})_{n\geq 1}$ is a sequence of functions. Note that for each point $c \in A$, $(f_n(c))n \geq 1$ will be a sequence of numbers. For example, consider $(f_n(x) = x^n)_{n\geq 1}$ on the set $A = [0, 2]$:

$$\begin{aligned} x = 0: &\quad f_1(0) = 0, f_2(0) = 0, f_3(0) = 0, ... \text{ is a sequence of numbers} \\ x = 1: &\quad f_1\left(\frac{1}{2}\right) = \frac{1}{2}, f_2\left(\frac{1}{2}\right) = \frac{1}{2^2}, f_3\left(\frac{1}{2}\right) = \frac{1}{2^3}, ... \text{ is a sequence of numbers} \\ x = 2: &\quad f_1(1) = 1, f_2(1) = 1, f_3(1) = 1, ... \text{ is a sequence of numbers} \\ &\quad \vdots \end{aligned}$$

Definition 3.1.2. (Pointwise Convergent Sequence)
We say a sequence of functions $(f_n : A \to \mathbb{R})_{n\geq 1}$ is pointwise convergent if at any $c \in A$, the sequence $(f_n(c))$ converges.

Example 3.1.2. For all $n \geq 1$, $f_n : \{0, 1\} \to \mathbb{R}, f_n(x) = 2x + \frac{x}{n}$.

$c \in \{0, 1\}$:	$(f_n(c))$	Convergence
0	$(f_n(0)) = (0, 0, 0, ...)$	$f_n(0) \to 0$
1	$(f_n(1)) = (2 + 1, 2 + \frac{1}{2}, 2 + \frac{1}{3}, ...)$	$f_n(1) \to 2$

So, (f_n) is a pointwise convergent sequence.

Definition 3.1.3. (Pointwise Limit)
Let $(f_n : A \to \mathbb{R})_{n\geq 1}$ be a pointwise convergent sequence. Then we define a new function on A as follows:

$$f : A \to \mathbb{R}, \quad \forall c \in A \;\; f(c) = \lim_{n\to\infty} f_n(c).$$

We call f the pointwise limit of the sequence (f_n) on A. We say f_n converges pointwise to the function f on A.

Example 3.1.3. For all $n \geq 1$, $f_n : [-1, 1] \to \mathbb{R}$, $f_n(x) = x^2 + \frac{1}{n}$. Prove that $f_n \to f$ pointwise on $[-1, 1]$ where $f(x) = x^2$.
At each point $c \in [-1, 1]$, the corresponding sequence of numbers

$$(f_n(c))_n \geq 1 = (c^2 + 1, c^2 + \frac{1}{2}, c^2 + \frac{1}{3}, ...) = (c^2 + \frac{1}{n})_{n\geq 1}.$$

For each $c \in [-1, 1]$, we have

$$\lim_{n\to\infty} f_n(c) = \lim_{n\to\infty} \left(c^2 + \frac{1}{n}\right) = c^2 = f(c).$$

Thus, $f_n \to f$ pointwise where $f : [-1, 1] \to \mathbb{R}$ defined by $f(x) = x^2$.

Example 3.1.4. For all $n \geq 1$, $f_n : [0, 1] \to \mathbb{R}$, $f_n(x) = \frac{x}{n}$. Provethat $f_n \to f$ where $f : [0, 1] \to \mathbb{R}$ is defined by $f(x) \equiv 0$.
At each point $c \in [0, 1]$, the corresponding sequence of numbers is

$$(f_n(c)) = \left(c, \frac{c}{2}, \frac{c}{3}, ...\right) = \left(\frac{c}{n}\right)_{n\geq 1}.$$

Thus for each $c \in [0, 1]$, we have

$$\lim_{n\to\infty} f_n(c) = \lim n \to \infty \frac{c}{n} = c \lim_{n\to\infty} \frac{1}{n} = c \cdot 0 = 0.$$

Example 3.1.5. $\forall n \geq 1$ $f_n : [0, 1] \to \mathbb{R}, f_n(x) = x^n$. Prove that $f_n \to f$ pointwise where $f : [0, 1] \to \mathbb{R}$ is defined by $f(x) = \begin{cases} 0 & \text{if } 0 \leq x < 1 \\ 1 & \text{if } x = 1 \end{cases}$.

At each point $c \in [0,1]$, the corresponding sequence of numbers is

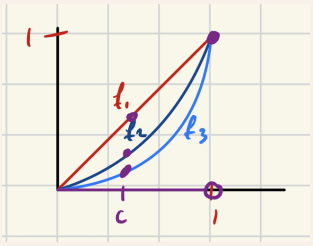
$$(f_c(c))_{n\geq 1} = (c^n)_{n\geq 1} = (c, c^2, c^3, ...).$$

Note that

$(*)$ if $0 \leq c < 1$, then $f_n(c) \to c^n = 0 = f(c)$.

$(*)$ if $c = 1$, then $f_n(c) \to 1 = f(c)$.

So, $f_n \to f$ pointwise.



Observation 3. In this example, each f_n is continuous but the pointwise limit f is not continuous.

Remark 3.1.2. The pointwise limit of a sequence of continuous functions is not necessarily continuous.

"Pointwise convergence does not preserve convergence."

Example 3.1.6. $\forall n \geq 1$ $f_n : [0,1] \to \mathbb{R}, f_n(x) = \begin{cases} x & \text{if } x \leq n \\ n & \text{if } x > n \end{cases}$. Prove $f_n \to f$ pointwise where $f : [0,\infty) \to \mathbb{R}$ is defined by $f(x) = x$.
At each point $c \in [0,\infty)$, the corresponding sequence of numbers is

$$(f_n(c))_{n\geq 1} = (f_1(c), f_2(c), f_3(c), ...).$$

Note that for fixed $c \in [0,\infty)$, $\exists N \in \mathbb{N}$ such that $c \leq N$, so clearly

$$c \leq N, \quad c \leq N+1, \quad c \leq N+2, ...$$

Thus

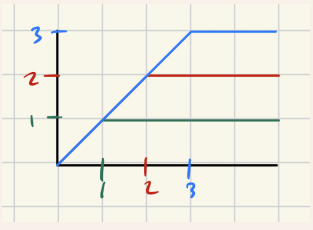
$$f_N(c) = c, f_{N+1}(c) = c, f_{N+2}(c) = c, ...$$

Therefore,

$$(f_n(c))_{n\geq 1} = (f_1(c), f_2(c), f_3(c), ..., f_{N-1}(c), c, c, c, ...).$$

We conclude that

$$\lim_{n\to\infty} f_n(c) = c.$$



Observation 4. Each f_n is bounded, but $f(x) = x$ is not bounded.

Remark 3.1.3. The pointwise limit of a sequence of bounded functions is not necessarily bounded.

"Pointwise convergence does not preserve boundedness."

Example 3.1.7. Let $f : [0,1] \to \mathbb{R}$ be a continuous, nowhere differentiable function. By the Weierstrass Approximation Theorem, there exists a sequence of polynomials $p_1(x), p_2(x), p_3(x), ...$ such that p_n converges to f pointwise.

Observation 5. Each p_n is differentiable, but the pointwise limit f is not differentiable.

Remark 3.1.4. The pointwise limit f of a sequence of differentiable functions is not necessarily differentiable. Also, if f' exists, it's not the case that $f_n' \to f'$.

"Pointwise convergence does not preserve differentiability."

Example 3.1.8. $\forall n \geq 4$, define $f_n : [0,1] \to \mathbb{R}$ as follows:

$$f_n(x) = \begin{cases} 0 & \text{if } 0 \leq x < \frac{1}{n} \\ n(nx-1) & \text{if } \frac{1}{n} \leq x < \frac{2}{n} \\ -n(nx-3) & \text{if } \frac{2}{n} \leq x < \frac{3}{n} \\ 0 & \text{if } \frac{3}{n} \leq x \leq 1 \end{cases}.$$

Prove that $f_n \to f$ pointwise where $f \equiv 0$.
At each fixed point $c \in [0,1]$, the corresponding sequence of numbers

$$(f_n(c))_{n\geq 4} = (f_4(c), f_5(c), f_6(c), ...).$$

Note that

$$\begin{aligned} c = 0: \quad (f_n(0))_{n\geq 4} = (0,0,0,0,...) &\implies \lim_{n\to\infty} f_n(0) = 0 \\ 0 < c \leq 1: \quad \exists N \in \mathbb{N} \text{ such that } \frac{3}{N} < c &\implies \frac{3}{N} < c, \frac{3}{N+1} < c, \frac{3}{N+2} < c, ... \\ &\implies f_N(c) = 0, f_{N+1}(c) = 0, f_{N+2}(c) = 0, ... \\ &\implies (f_n(c))_{n\geq 4} = (f_4(c), f_5(c), ..., f_{N-1}(c), 0, 0, ...) \\ &\implies \lim_{n\to\infty} f_n(c) = 0. \end{aligned}$$

$\forall c \in [0,1]$, $\lim_{n\to\infty} f_n(c) = 0 = f(c)$, so $f_n \to f = 0$ pointwise.

Observation 6. Each f_n, $\int_0^n f_n = 1$. However, $\int_0^1 f = 0$, so for every n $\int_0^1 f_n \neq \int_0^1 f$.

Remark 3.1.5. Pointwise convergence does not allow us to interchange limits and integrals.

"Pointwise convergence does not preserve integrals."

3.2 Uniform Convergence

Definition 3.2.1. (Uniform Convergence)
We say $(f_n : A \to \mathbb{R})$ converges uniformly to $f : A \to \mathbb{R}$ if

$$\forall \epsilon > 0 \;\; \exists N_\epsilon \text{ such that } \forall n > N_\epsilon \; \forall c \in A \;\; |f_n(c) - f(c)| < \epsilon.$$

Notation . $f_n \overset{u}{\to} f$.

Example 3.2.1. $\forall n \geq 1 \;\; f_n : \mathbb{R} \to \mathbb{R}, \; f_n(x) = \dfrac{1}{n(1+x^2)}$.

(i) Prove that $f_n \to f$ pointwise where $f : \mathbb{R} \to \mathbb{R}$ is defined by $f = 0$.
At each point $c \in \mathbb{R}$, the corresponding sequence of real numbers is

$$(f_n(c))_{n\geq 1} = \left(\frac{1}{n(1+c^2)}\right)_{n\geq 1} = \left(\frac{1}{1+c^2}, \frac{1}{2(1+c^2)}, \frac{1}{3(1+c^2)}, \ldots\right).$$

Thus for each $c \in \mathbb{R}$ we have

$$\lim_{n\to\infty} f_n(c) = \lim_{n\to\infty}\left[\frac{1}{n(1+c^2)}\right] = \frac{1}{1+c^2}\left(\lim_{n\to\infty}\frac{1}{n}\right) = 0 = f(c).$$

So, $f_n \to f$ pointwise where $f \equiv 0$.

(ii) Prove that $f_n \overset{u}{\to} f$.
Our goal is to show that

$$\forall \epsilon > 0 \;\; \exists N \text{ such that } \forall n > N \; \forall c \in \mathbb{R} \; |f_n(c) - 0| < \epsilon.$$

Let $\epsilon > 0$ be given. Our goal is to find N such that

$$\forall n > N \;\; \forall c \in \mathbb{R} \; \frac{1}{n(1+c^2)} < \epsilon.$$

By Archimedean property, $\exists \hat{N}$ such that $\frac{1}{\hat{N}} < \epsilon$. Then $\forall n > \hat{N}$ and $\forall c \in \mathbb{R}$, we have

$$\frac{1}{n(1+c^2)} \leq \frac{1}{n} < \frac{1}{\hat{N}} < \epsilon$$

as desired.

Example 3.2.2. $\forall n \geq 1$, $f_n : [0,1] \to \mathbb{R}$, $f_n(x) = \frac{x}{n}$. Prove that $f_n \to 0$ uniformly on $[0,1]$.
Our goal is to the show that

$$\forall \epsilon > 0 \;\; \exists N \text{ such that } \forall n > N \;\; \forall c \in [0,1] \;\; |f_n(0) - 0| < \epsilon.$$

Let $\epsilon > 0$ be given. By the Archimedean Property, $\exists \hat{N}$ such that $\frac{1}{\hat{N}} < \epsilon$. We claim that this $\hat{N}$ can be used as the N that we are looking for. Indeed, $\forall n > \hat{N}$ and $\forall c \in [0,1]$ we have

$$\frac{c}{n} \leq \frac{1}{n} < \frac{1}{N} < \epsilon$$

as desired.

Example 3.2.3. $\forall n \geq 1$, $f_n : [0,\infty) \to \mathbb{R}$, $f_n(x) = \frac{x}{n}$.

(i) Prove that $f_n \to 0$ pointwise on $[0,\infty)$.
At each point $c \in [0,\infty)$, the corresponding sequence of real numbers is

$$(f_n(c))_{n\geq 1} = \left(\frac{c}{n}\right)_{n\geq 1} = \left(c, \frac{c}{2}, \frac{c}{3}, \frac{c}{4}, \ldots\right).$$

Thus, for each $c \in [0, \infty)$,

$$\lim_{n\to\infty} f_n(c) = \lim_{n\to\infty} \frac{c}{n} = c\left(\lim_{n\to\infty} \frac{1}{n}\right) = 0.$$

So, $f_n \to f$ pointwise where $f \equiv 0$.

(ii) Prove that the convergence is not uniform on $[0, \infty)$.

Assume for contradiction that $f_n \overset{u}{\to} 0$ on $[0, \infty)$. Then

$$\forall \epsilon > 0 \;\; \exists N \text{ such that } \forall n > N \; \forall c \in [0, \infty) \;\; |f_n(c) - 0| < \epsilon.$$

In particular, for $\epsilon = 1$, $\exists N \in \mathbb{N}$ such that

$$\forall n > N \; \forall c \in [0, \infty) \;\; \frac{c}{n} < 1.$$

Letting $n = N + 1$, we get

$$\forall c \in [0, \infty) \;\; \frac{c}{N+1} < 1.$$

Thus,

$$\forall c \in [0, \infty) \;\; c < N + 1.$$

This is clearly false.

Theorem 3.2.1. (Uniform Converges Preserves Boundedness)
Suppose A is not empty, and for each $n \in \mathbb{N}$ let $f_n : A \to \mathbb{R}$ be bounded, and $f_n \overset{u}{\to} f$ on A. Then $f : A \to \mathbb{R}$ is bounded.

Proof. Our goal is to show that

$$\exists M \in \mathbb{R} \quad \text{such that } \forall c \in A \;\; |f(c)| \leq M.$$

Since $f_n \overset{u}{\to} f$ on A, we have

$$\begin{aligned} &\forall \epsilon > 0 \;\; \exists N \text{ such that } \forall n > N \; \forall c \in A \;\; |f_n(c) - f(c)| < \epsilon \\ \implies &\forall \epsilon > 0 \;\; \exists N \text{ such that } \forall n > N \; \forall c \in A \;\; |f_n(c)| - |f(c)| < \epsilon \\ \implies &\forall \epsilon > 0 \;\; \exists N \text{ such that } \forall n > N \; \forall c \in A \;\; |f(c)| < \epsilon + |f_n(c)|. \end{aligned}$$

In particular, for $\epsilon = 1$, $\exists N \in \mathbb{N}$ such that

$$\forall n > N \; \forall c \in A \;\; |f(c)| < 1 + |f_n(c)|.$$

Letting $n = N + 1$, we get

$$\forall c \in A \;\; |f(c)| < 1 + |f_{N+1}(c)|. \tag{1}$$

Since, by assumption, f_{N+1} is bounded, $\exists \hat{M}_{N+1}$ such that

$$\forall c \in A \;\; |f_{N+1}(c)| \leq \hat{M}_{N+1}. \tag{2}$$

It follows from (1) and (2) that

$$\forall c \in A \;\; |f(c)| < 1 + \hat{M}_{N+1}.$$

Clearly, we can use $1 + \hat{M}_{N+1}$ as the M we were looking for. □

Theorem 3.2.2. (Uniform Convergence Preserves Continuity)
Let (X, d) be a metric space. Suppose $A \subseteq (X, d)$, $c \in A$, for each $n \in \mathbb{N}$ $f_n : A \to R$ is continuous at c, and $f_n \overset{u}{\to} f$ on A. Then $f : A \to \mathbb{R}$ is continuous at c.

Proof. Our goal is to show that

$$\forall \epsilon > 0 \;\; \exists \delta > 0 \text{ such that } \text{ if } d(x, c) < \delta \text{ for } x \in A, \text{ then } |f(x) - f(c)| < \epsilon.$$

Let $\epsilon > 0$ be given. Our goal is to find $\delta > 0$ satisfying

$$\text{if } d(x,c) < \delta, \text{ then } |f(x) - f(c)| < \epsilon.$$

Informal Discussion

Recall that our general strategy to ensure that a quantity of interest is less than ϵ is as follows:

Step 1: Bound quantity 1 with quantity 2, which is a combination of terms that are easier to deal with or are terms over which we can control.

Step 2: Make sure quantity 2 is less than ϵ.

According to the hypothesis of this theorem, here is a list of quantities over which we have control:

(i) $f_n \overset{u}{\to} f$ on $A \implies$ we can make $|f_n(x) - f(x)|$ small $\forall x$.

(ii) f_n is continuous at $c \implies$ we can make $|f_n(x) - f(c)|$ small for each n.

We have

$$\begin{aligned} |f(x) - f(c)| &= |f(x) - f_n(x) + f_n(x) - f_n(c) + f_n(c) - f(c)| \\ &\leq |f(x) - f_n(x)| + |f_n(x) - f_n(c)| + |f_n(c) - f(c)|. \end{aligned}$$

So, in order to ensure $|f(x) - f(c)| < \epsilon$, it is enough to make each piece on the RHS less than $\frac{\epsilon}{3}$. We will first find f_n that forces the first term and the third term to be less than $\frac{\epsilon}{3}$. Once we establish which f_n we want, we can use the continuity of f_n at c to produce a δ that makes the middle term be less than $\frac{\epsilon}{3}$ for $x \in N_\delta(c) \cap A$.

Since $f_n \overset{u}{\to} f$ on A,

$$\exists N \in \mathbb{N} \text{ such that } \forall n > N \ \forall z \in A \ \ |f_n(z) - f(z)| < \frac{\epsilon}{3}. \tag{1}$$

Also, since f_{N+1} is continuous at c,

$$\exists \hat{\delta} > 0 \text{ such that } \forall x \in N_{\hat{\delta}}(c) \cap A \ \ |f_{N+1}(x) - f_{N+1}(c)| < \frac{\epsilon}{3}. \tag{2}$$

We claim that this $\hat{\delta}$ can be used as the δ we are looking for. For all $x \in N_\delta(c) \cap A$, we have

$$\begin{aligned} |f(x) - f(c)| &= |f(x) - f_{N+1}(x) + f_{N+1}(x) - f_{N+1}(c) + f_{N+1}(c) - f(c)| \\ &\leq |f(x) - f_{N+1}(x)| + |f_{N+1}(x) - f_{N+1}(c)| + |f_{N+1}(c) - f(c)| \\ &< \frac{\epsilon}{3} + \frac{\epsilon}{3} + \frac{\epsilon}{3} \qquad ((1) \text{ and } (2)) \\ &= \epsilon \end{aligned}$$

as desired. □

Observation 7. Let (a_n) be a sequence of real numbers. Suppose $a_n \to a \in \mathbb{R}$. Suppose there exists N such that

$$\forall m, n > N \ \ |a_n - a_m| \leq \frac{1}{3}.$$

Note that, for each fixed $n > N$, we have

$$\forall m > N \ \ |a_n - a_m| \leq \frac{1}{3}.$$

So, by taking the limit as $m \to \infty$, it follows from the Order Limit Theorem that for each $n > N$

$$\lim_{m\to\infty} |a_n - a_m| \leq \lim_{m\to infty} \frac{1}{3}.$$

Therefore,

$$\forall n > N \ \ |a_n - a| \leq \frac{1}{3}.$$

More generally, given $\epsilon > 0$, if there exists N such that

$$\forall n, m > N \ \ |a_n - a_m| \leq \epsilon$$

then by taking the limit as $m \to \infty$, we obtain

$$\forall n > N \quad |a_n - a| \leq \epsilon.$$

Theorem 3.2.3. (Cauchy Criterion for Uniform Convergence)
Let $A \neq \emptyset$. Suppose for each $n \in \mathbb{N}$, $f_n : A \to \mathbb{R}$ is a function. Then

$$(f_n)_{n\geq 1} \text{ converges uniformly } \iff \forall \epsilon > 0 \ \exists N \text{ such that } \forall n, m > N \ \forall c \in A \quad |f_n(c) - f_m(c)| < \epsilon.$$

Proof. ($\implies$) Suppose $\exists f : A \to \mathbb{R}$ such that $f_n \overset{u}{\to} f$ on A. Let $\epsilon > 0$ be given. Our goal is to find N such that

$$\forall n, m > N \ \forall c \in A \quad |f_n(c) - f_m(c)| < \epsilon.$$

Since $f_n \overset{u}{\to} f$ on A, for the given $\epsilon > 0$ $\exists \hat{N}$ such that

$$\forall k > \hat{N} \ \forall c \in A \quad |f_k(c) - f(c)| < \frac{\epsilon}{2}.$$

We claim this $\hat{N}$ can be used as the N we're looking for. Indeed, if $m, n > \hat{N}$ and $c \in A$ then

$$|f_n(c) - f_m(c)| \leq |f_n(c) - f(c)| + |f_m(c) - f(c)| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.$$

($\impliedby$) Suppose $\forall \epsilon > 0$ $\exists N$ such that $\forall n, m > N$ $\forall c \in A$ $|f_n(c) - f_m(c)| < \epsilon$. It follows from the assumption that at each point $c \in A$, the sequence of real numbers $(f_n(c))_{n\geq 1}$ is a Cauchy sequence in $\mathbb{R}$. Since $\mathbb{R}$ is complete, we can conclude that for each point $c \in A$, the sequnce of real numbers $(f_n(c))_{n\geq 1}$ converges. This tells us that the sequence of functions $(f_n)_{n\geq 1}$ is pointwise convergent on A. Let's denote the pointwise limit of $(f_n)_{n\geq 1}$ by $f : A \to \mathbb{R}$. In what follows, we will prove $f_n \overset{u}{\to} f$ on A. To this end, we need to show

$$\forall n > N \ \forall c \in A \quad |f_n(c) - f(c)| < \epsilon.$$

It follows from the assumption that for the given $\epsilon > 0$, $\exists \hat{N}$ such that

$$\forall n, m > \hat{N} \ \forall c \in A \quad |f_n(c) - f_m(c)| < \frac{\epsilon}{2}.$$

Then, if $n > \hat{N}$ and $c \in A$ we have

$$\begin{aligned} &\forall m > \hat{N} \quad |f_n(c) - f_m(c)| < \frac{\epsilon}{2} \\ \implies &\lim_{n\to\infty} |f_n(c) - f_m(c)| \leq \lim_{n\to\infty} \frac{\epsilon}{2} \\ \implies &|f_n(c) - f(c)| \leq \frac{\epsilon}{2} < \epsilon. \end{aligned}$$

□

Theorem 3.2.4. (Uniform Continuity and Differentiability)
Suppose for each $n \in \mathbb{N}$, $f_n : [a, b] \to \mathbb{R}$ is differentiable, $f_n \to f$ pointwise on $[a, b]$, and $f_n' \overset{u}{\to} g$ on $[a, b]$. Then f is differentiable on $[a, b]$ and $f' = g$.

Proof. Our goal is to show that $\forall c \in [a, b]$, $f'(c) = g(c)$. Let $c \in [a, b]$. We want to show

$$\lim_{x\to c} \frac{f(x) - f(c)}{x - c} = g(c).$$

That is, we want to show

$$\forall \epsilon > 0 \ \exists \delta > 0 \text{ such that if } 0 < \underset{x\in[a,b]}{|x - c|} < \delta, \text{ then } \left| \frac{f(x) - f(c)}{x - c} - g(c) \right| < \epsilon. \qquad (*)$$

Let $\epsilon > 0$ be given.

Informal Discussion

Strategy: Bound $\left|\frac{f(x)-f(c)}{x-c} - g(c)\right|$ with a combintation of terms over which we control.

$*)$ f_n is differentiable $\implies$ we control $\left|\frac{f_n(x)-f_n(c)}{x-c} - f_n'(c)\right|$.

$*)$ $f_n' \overset{u}{\to} g \implies$ we control $|f_n'(z) - g(z)|$ for all $z \in [a,b]$.

$*)$ $f_n \to f$ pointwise $\implies$ we control $|f_n(z) - f(z)|$ at each $z \in [a,b]$.

Note that:

$$\begin{aligned}\left|\frac{f(x)-f(c)}{x-c} - g(c)\right| &= \left|\frac{f(x)-f(c)}{x-c} - \frac{f_n(x)-f_n(c)}{x-c} + \frac{f_n(x)-f_n(c)}{x-c} - f_n'(c) + f_n'(c) - g(c)\right| \\ &\le \left|\frac{f(x)-f(c)}{x-c} - \frac{f_n(x)-f_n(c)}{x-c}\right| + \left|\frac{f_n(x)-f_n(c)}{x-c} - f_n'(c)\right| + |f_n'(c) - g(c)|.\end{aligned}$$

In order to ensure $\left|\frac{f(x)-f(c)}{x-c} - g(c)\right| < \epsilon$, it's enough to make each term less than $\epsilon/3$. We will find an f_n that forces the first and third term to be less than $\epsilon/3$. We use the differentiability of f_n at c to find a δ that makes the middle term less than $\epsilon/3$ for $x \in [a,b]$ with $0 < |x-c| < \delta$.

(1) Since $f_n' \overset{u}{\to} g$, $\exists N_1$ such that $\forall n > N_1 \quad \forall z \in [a,b] \quad |f_n'(z) - g(z)| < \epsilon/3$.
(2) Since $f_n' \overset{u}{\to} g$, by the Cauchy Criterion $\exists N_2$ such that $\forall n, m > N_2 \quad \forall z \in [a,b] \quad |f_n'(z) - f_m'(z)| < \epsilon/3$.
Let $N = \max\{N_1, N_2\} + 1$.
(3) f_n is differentiable at c, so for the given ϵ

$$\exists \hat{\delta} > 0 \text{ such that if } x \in [a,b] \text{ and } 0 < |x-c| < \hat{\delta}, \text{ then } \left|\frac{f_N(x)-f_N(c)}{x-c} - f_N'(c)\right| < \epsilon/3.$$

We claim this $\hat{\delta}$ can be used as to satisfy $(*)$. Indeed, if $x \in [a,b]$ and $0 < |x-c| < \hat{\delta}$, then

$$\begin{aligned}\left|\frac{f(x)-f(c)}{x-c} - g(c)\right| &\le \left|\frac{f(x)-f(c)}{x-c} - \frac{f_N(x)-f_N(c)}{x-c}\right| + \left|\frac{f_N(x)-f_N(c)}{x-c} - f_N'(c)\right| + |f_N'(c) - g(c)| \\ &< \left|\frac{(f-f_N)(x)-(f-f_N)(c)}{x-c}\right| + \epsilon/3 + \epsilon/3. \qquad ((1) \text{ and } (3))\end{aligned}$$

We consider two cases: $x < c$ and $x > c$.

Case 1: $x \in [a,b]$, $x < c$.
For every $m > N$, apply the Mean Value Theorem to $f_m - f_N$ on $[x,c]$:

$$\begin{aligned}\forall m > N \ \exists \alpha_m \in [x,c] \text{ such that } (f_m - f_N)'(\alpha_m) &= \frac{(f_m-f_N)(c)-(f_m-f_N)(x)}{c-x} \\ \implies \left|\frac{(f_m-f_N)(c)-(f_m-f_N)(x)}{c-x}\right| &= |f_m'(\alpha_m) - f_N'(\alpha_m)| \\ &< \epsilon/3. \qquad ((2))\end{aligned}$$

By the order limit theorem with $m \to \infty$, we have

$$\begin{aligned}&\left|\frac{(fm-f_N)(c)-(fm-f_N)(x)}{c-x}\right| \le \epsilon/3 \\ \implies &\left|\frac{(f-f_N)(c)-(f-f_N)(x)}{c-x}\right| \le \epsilon/3,\end{aligned}$$

as desired.

Case 2: $x \in [a,b]$, $x > c$. Analogous to case 1.

□

Lemma 3.2.1. Let $A \neq \emptyset$ and. $f : A \to \mathbb{R}$. Suppose $(f_n : A \to \mathbb{R})$ is a sequence of functions. The following are equivalent:

(i) $f_n \overset{u}{\to} f$ on A.

(ii) $\lim_{n\to\infty} \sup_{x\in A} |f_n(x) - f(x)| = 0$.

Lemma 3.2.2. Suppose $\alpha : [a, b] \to \mathbb{R}$ is increasing, $f : [a, b] \to \mathbb{R}$ and $g : [a, b] \to \mathbb{R}$ are bounded, and $f \leq g$. Then

$$L(f, \alpha) \leq L(g, \alpha)$$
$$U(f, \alpha) \leq U(g, \alpha)$$

Theorem 3.2.5. (Uniform Continuity and Integrability)
Suppose $\alpha : [a, b] \to \mathbb{R}$ is increasing. For each $n \in \mathbb{N}$, let $f_n \in R_\alpha[a, b]$ with $f_n \overset{u}{\to} f$ on $[a, b]$. Then $f \in R_\alpha[a, b]$ and $\int_a^b f d\alpha = \lim_{n\to\infty} \int_a^b f_n d\alpha$.

Proof. Since uniform convergence preserves boundedness, we can conclude that $f : [a, b] \to \mathbb{R}$ is bounded. Now, in order to show that $f \in R_\alpha[a, b]$, it is enough to prove that $L(f, \alpha) = U(f, \alpha)$. For each $n \in \mathbb{N}$, let

$$r_n = x \in \sup_{[a,b]} |f_n(x) - f(x)|.$$

Since $f_n \to f$ uniformly, we know that $\lim_{n\to\infty} r_n = 0$. For each $n \in \mathbb{N}$, we have

$$r_n = \sup_{x\in[a,b]} |f_n(x) - f(x)| \implies \forall x \in [a, b] \;\; |f(x) - f_n(x)| \leq r_n$$
$$\implies \forall x \in [a, b] \;\; f_n(x) - r_n \leq f(x) \leq f_n(x) + r_n. \qquad (*)$$

It follows from lemma 3.2.2 that

$$L(f_n - r_n, \alpha) \leq L(f, \alpha) \leq U(f, \alpha) \leq U(f_n + r_n, \alpha).$$

Thus

$$0 \leq U(f, \alpha) - L(f, \alpha) \leq U(f_n + r_n, \alpha) - L(f_n - r_n, \alpha).$$

Note that

$$\begin{aligned} U(f_n + r_n, \alpha) - L(f_n - r_n, \alpha) &= \int_a^b f_n + r_n d\alpha - \int_a^b f_n - r_n d\alpha \qquad (f_n\text{'s are integrable}) \\ &= \int_a^b (f_n + r_n) - (f_n - r_n) d\alpha \\ &= 2r_n \int_a^b d\alpha \\ &= 2r_n[\alpha(b) - \alpha(b)]. \end{aligned}$$

So,

$$0 \leq U(f, \alpha) - L(f, \alpha) \leq 2r_n[\alpha(b) - \alpha(a)] \to 0[\alpha(b) - \alpha(a)] = 0 \text{ as } n \to \infty.$$

By the squeeze theorem, $U(f, \alpha) = L(f, \alpha)$. From $(*)$,

$$\int_a^b f_n - r_n d\alpha \leq \int_a^b f d\alpha \leq \int_a^b f_n + r_n d\alpha,$$

so

$$\int_a^b -r_n d\alpha \leq \int_a^b f d\alpha - \int_a^b f_n d\alpha \leq \int_a^b r_n d\alpha.$$

Thus,

$$-r_n[\alpha(b) - \alpha(a)] \leq \int_a^b f d\alpha - \int_a^b f_n d\alpha \leq r_n[\alpha(b) - \alpha(a)].$$

By the squeeze theorem, $\lim_{n\to\infty} \left[\int_a^b f d\alpha - \int_a^b f_n d\alpha\right] = 0$. That is,

$$\lim_{n\to\infty} \int_a^b f_n d\alpha = \int_a^b f d\alpha$$

□